



US 20250286647A1

(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0286647 A1**
(43) **Pub. Date: Sep. 11, 2025**(54) **MODULATION SCHEMES FOR NONBINARY POLAR CODING**(52) **U.S. Cl.**CPC *H04L 1/0003* (2013.01); *H03M 13/13* (2013.01); *H04L 1/0057* (2013.01)(71) Applicant: **QUALCOMM Incorporated**, San Diego, CA (US)

(57)

ABSTRACT

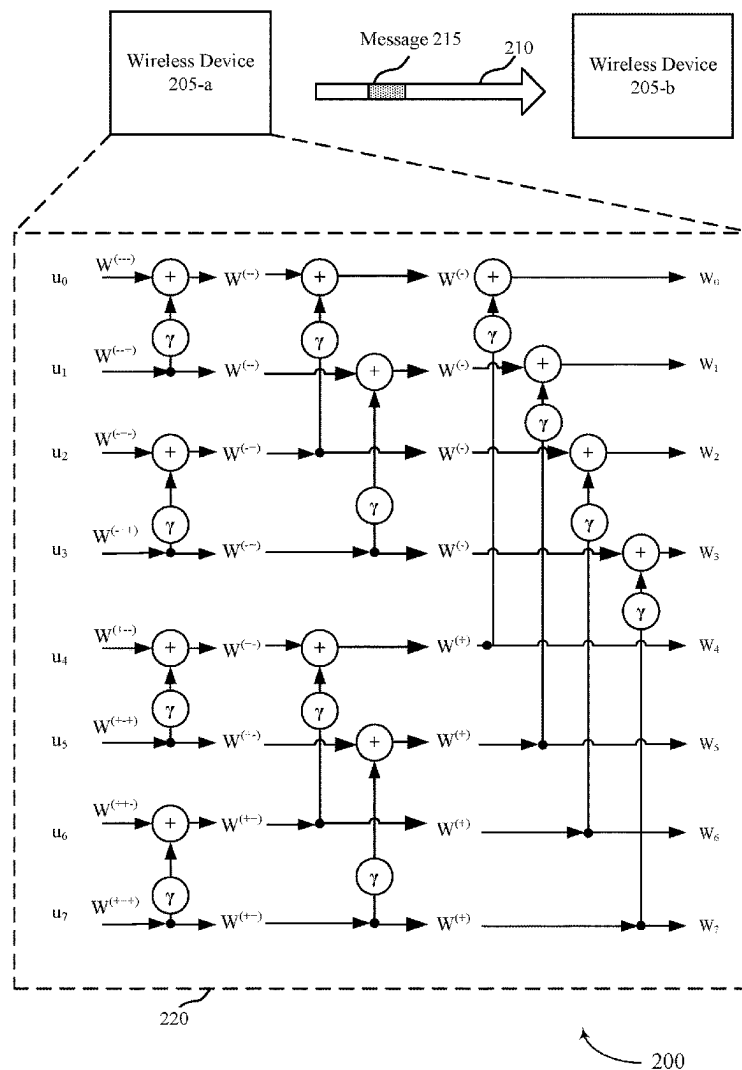
Methods, systems, and devices for wireless communications are described. The described techniques relate to improved methods, systems, devices, and apparatuses that support modulation schemes for nonbinary polar coding. For example, the described techniques may enable a transmitting device to perform a modulation of a set of symbols (e.g., bit pairs) that are encoded using a nonbinary polar encoding scheme. For example, the transmitting device may map each symbol of a set of encoded symbols to a constellation point. Additionally, or alternatively, the transmitting device may convert the encoded symbols into strings of bits. The transmitting device may accordingly perform bit-to-constellation mapping of the resulting bitstring using an existing mapping scheme or by defining a new mapping scheme. A receiving device may demodulate a received message by performing a demodulation of the set of modulated symbols or bits (e.g., and by converting the bitstring into the set of information symbols).

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South Orange, NJ (US)(21) Appl. No.: **18/596,006**(22) Filed: **Mar. 5, 2024****Publication Classification**(51) **Int. Cl.***H04L 1/00*

(2006.01)

H03M 13/13

(2006.01)



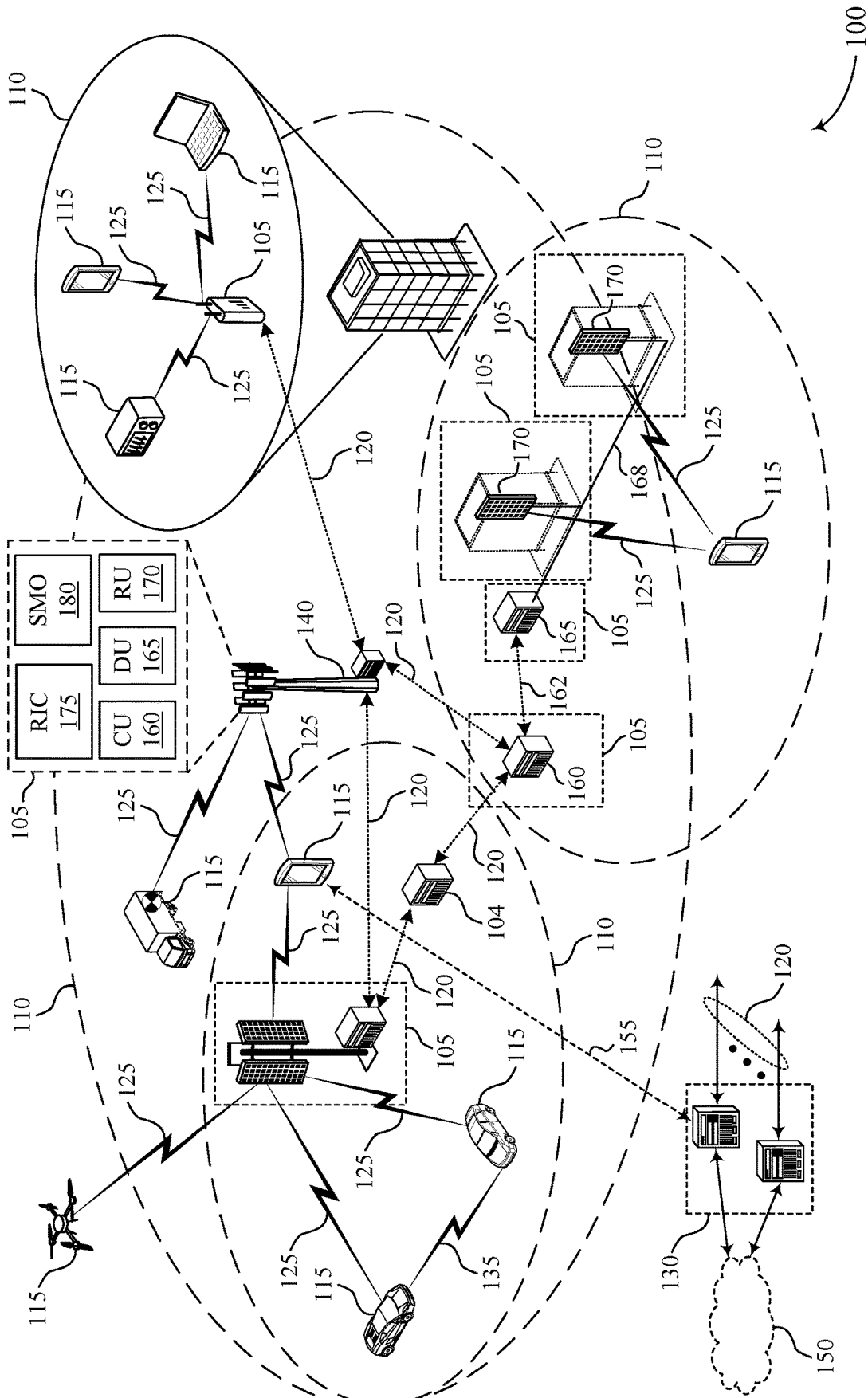


FIG. 1

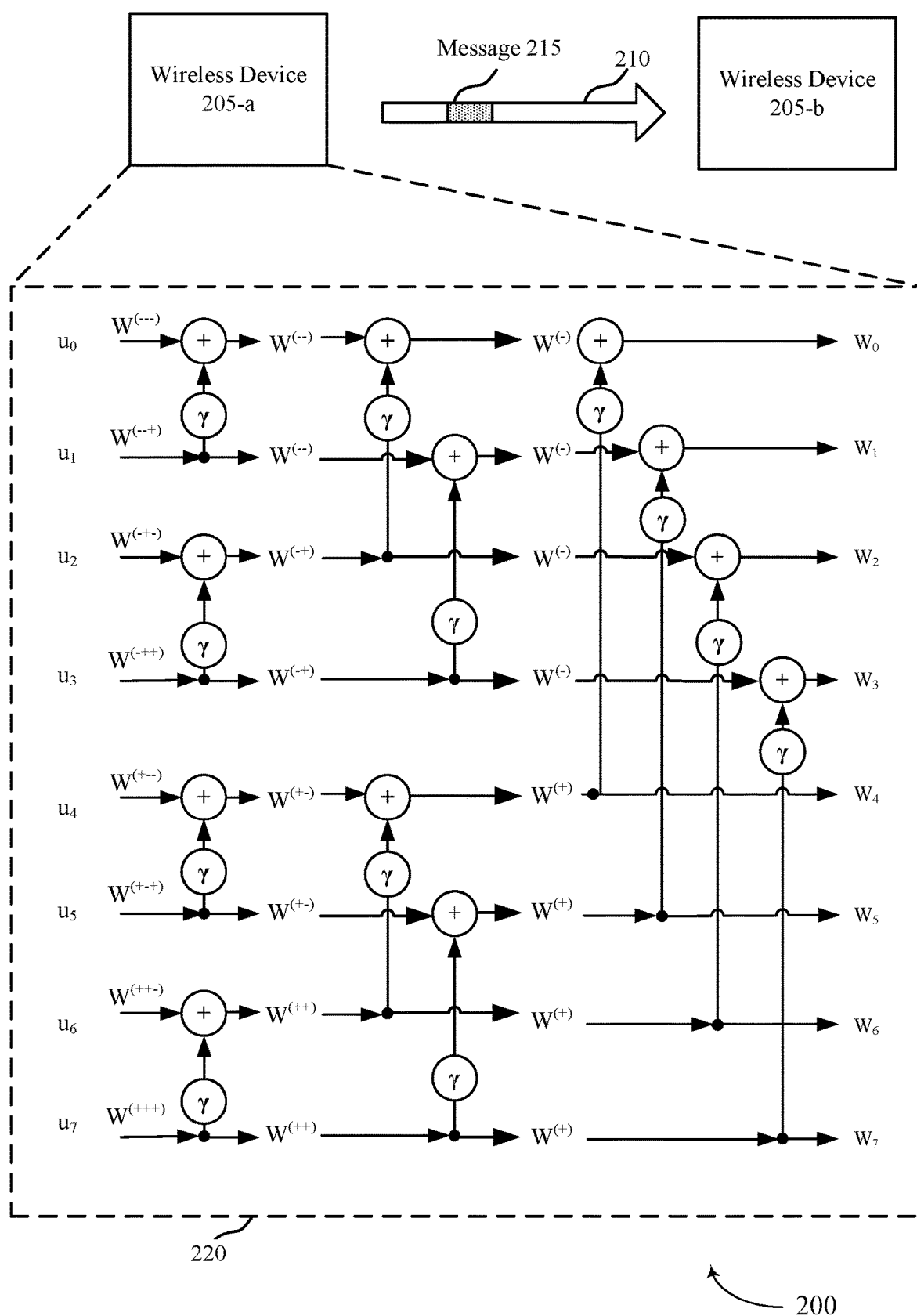


FIG. 2

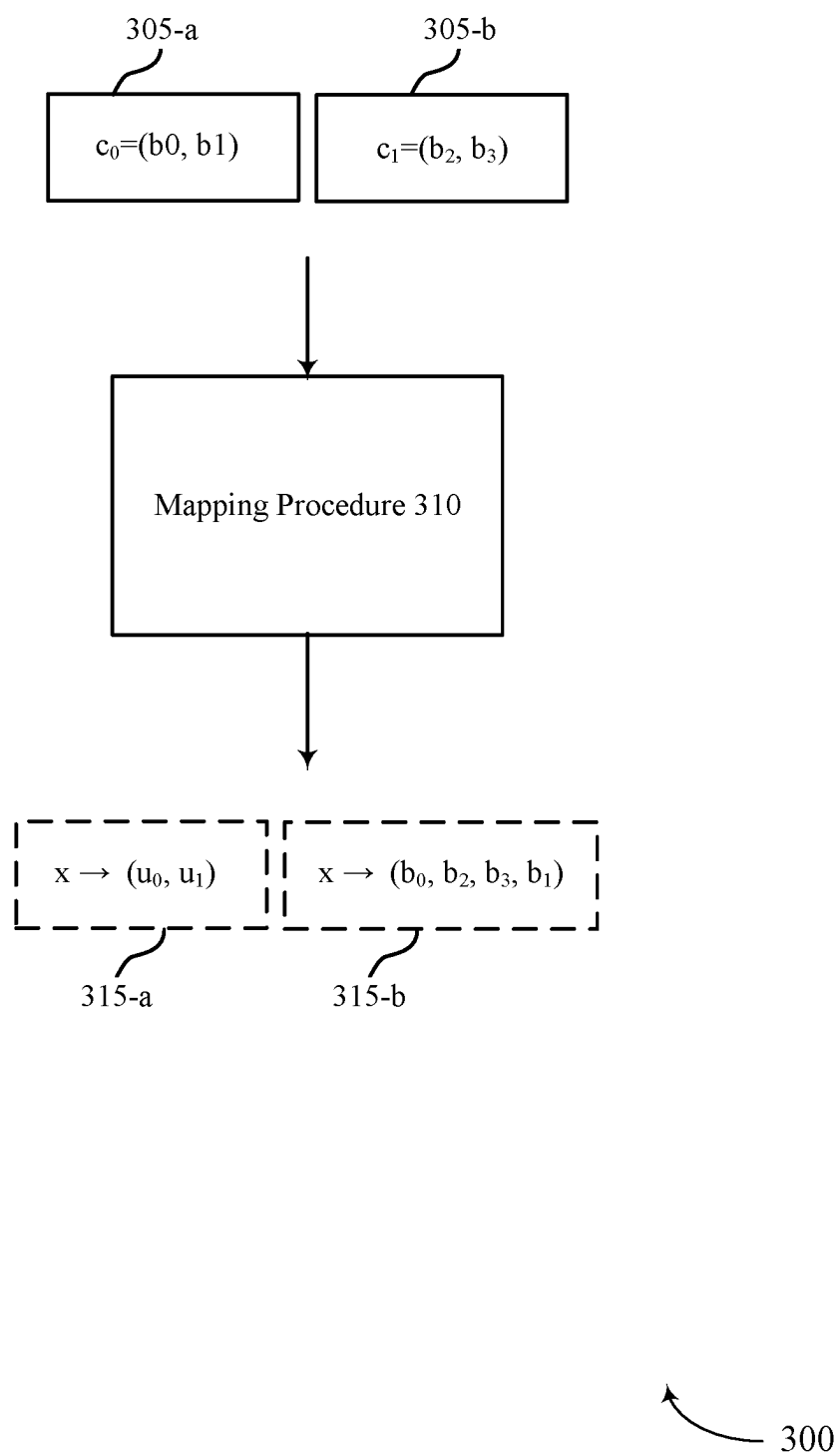


FIG. 3

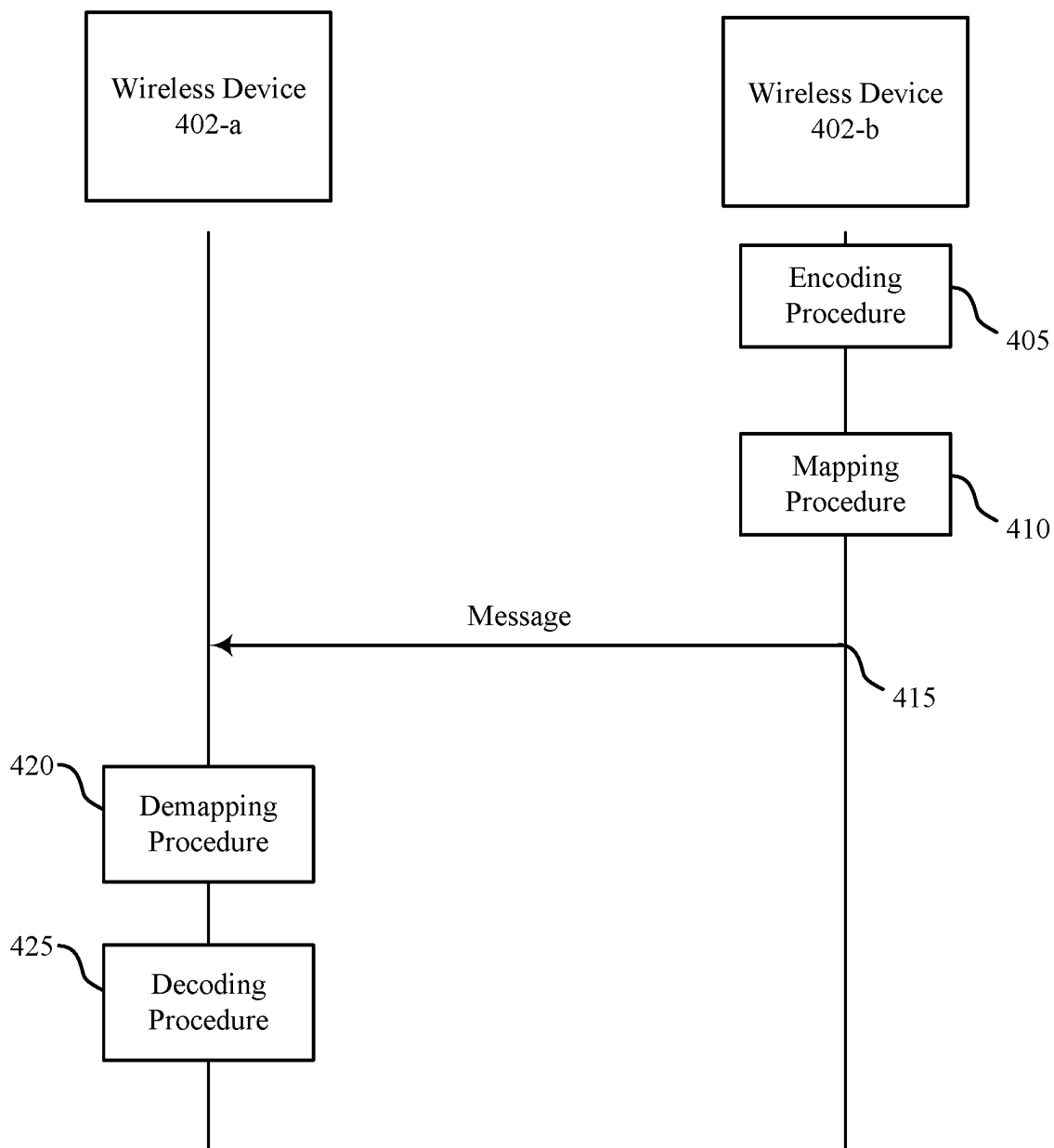
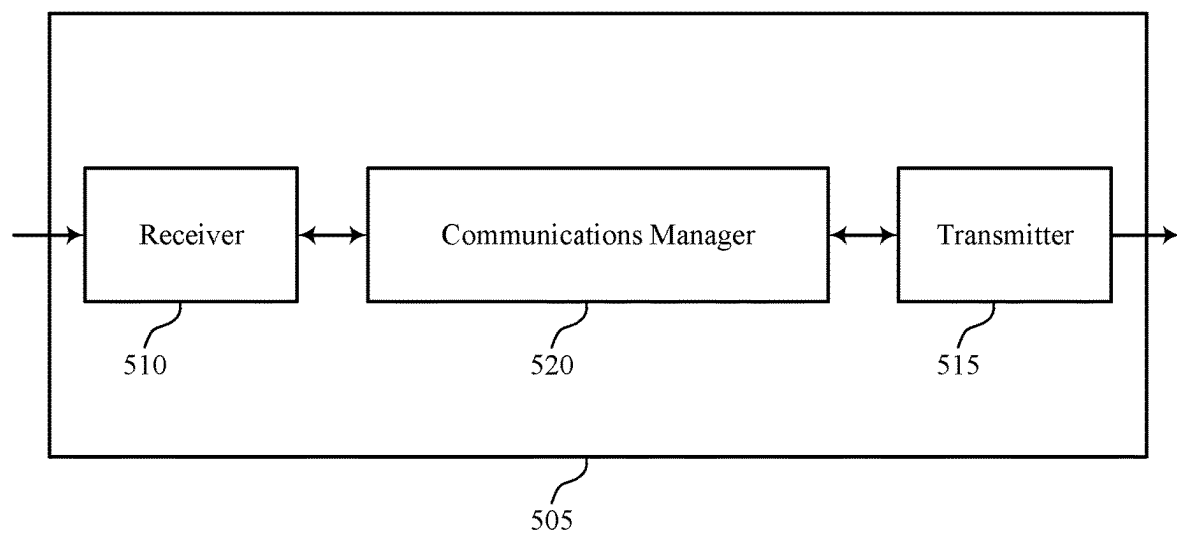


FIG. 4



500

FIG. 5

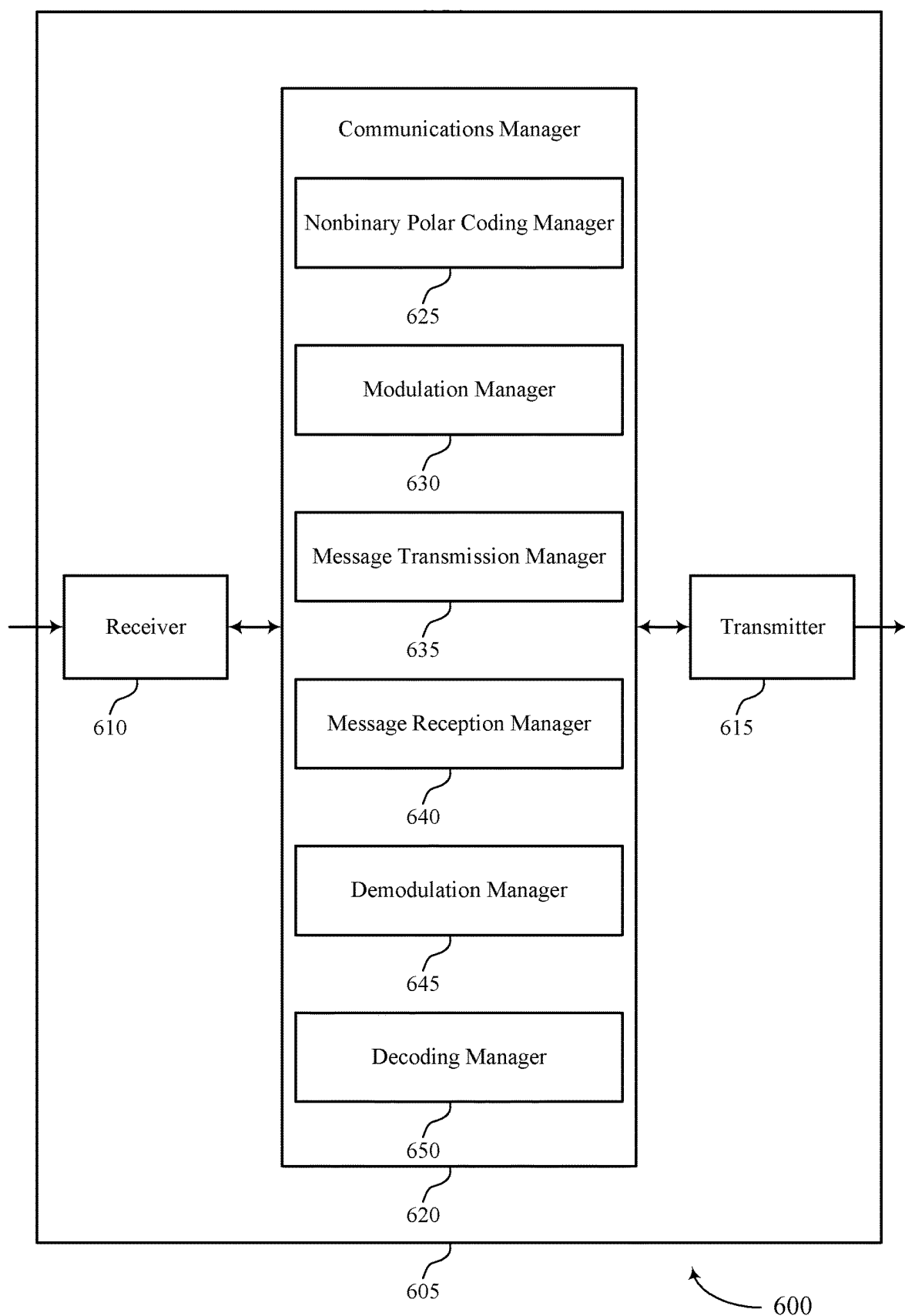


FIG. 6

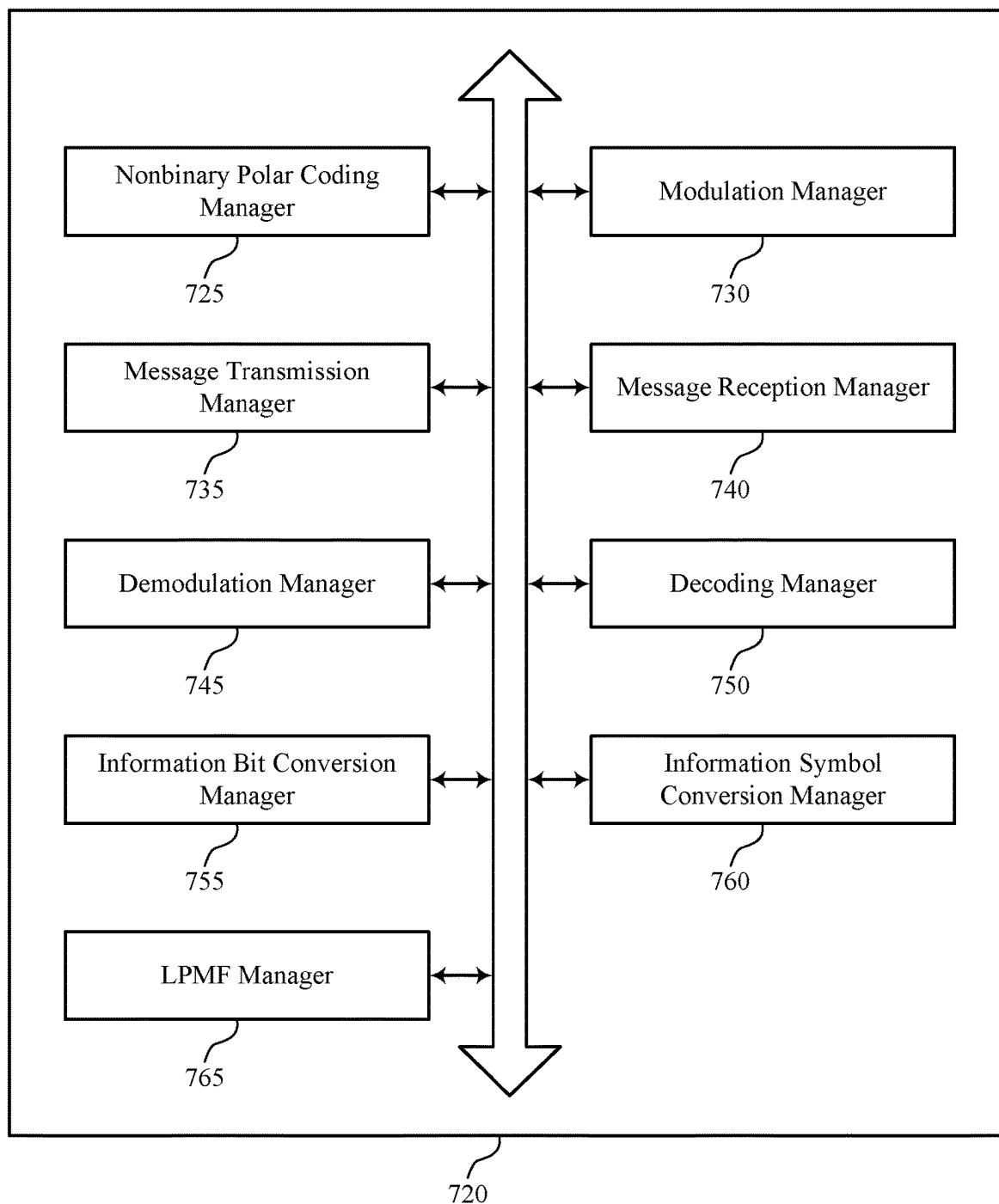


FIG. 7

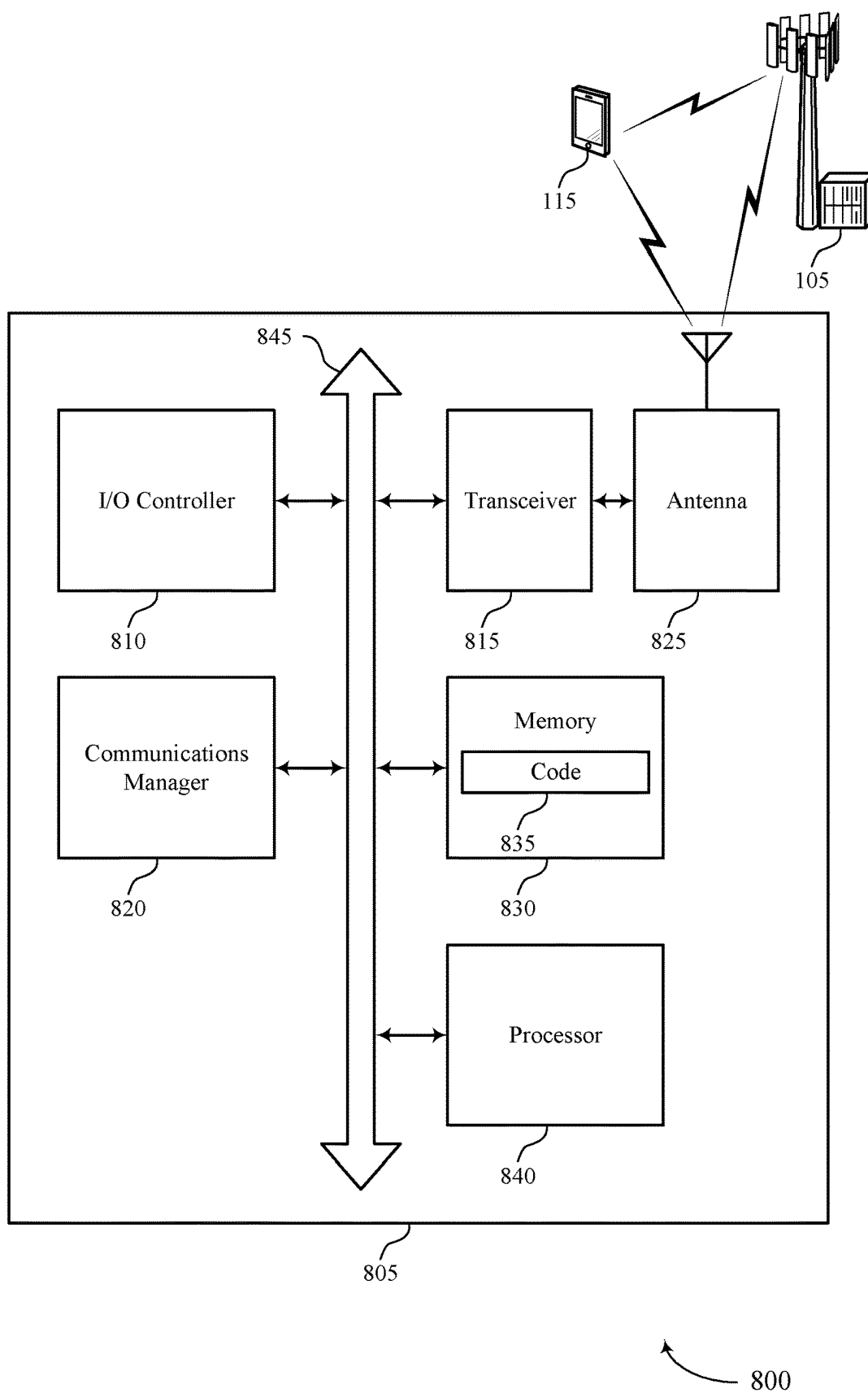


FIG. 8

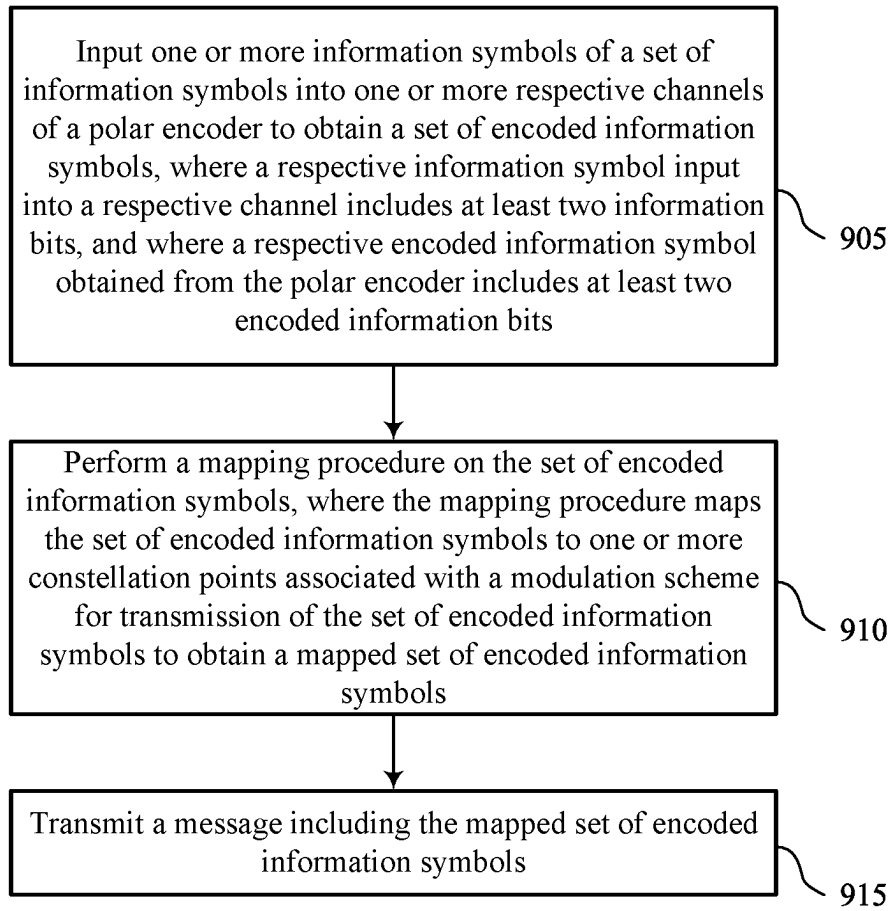
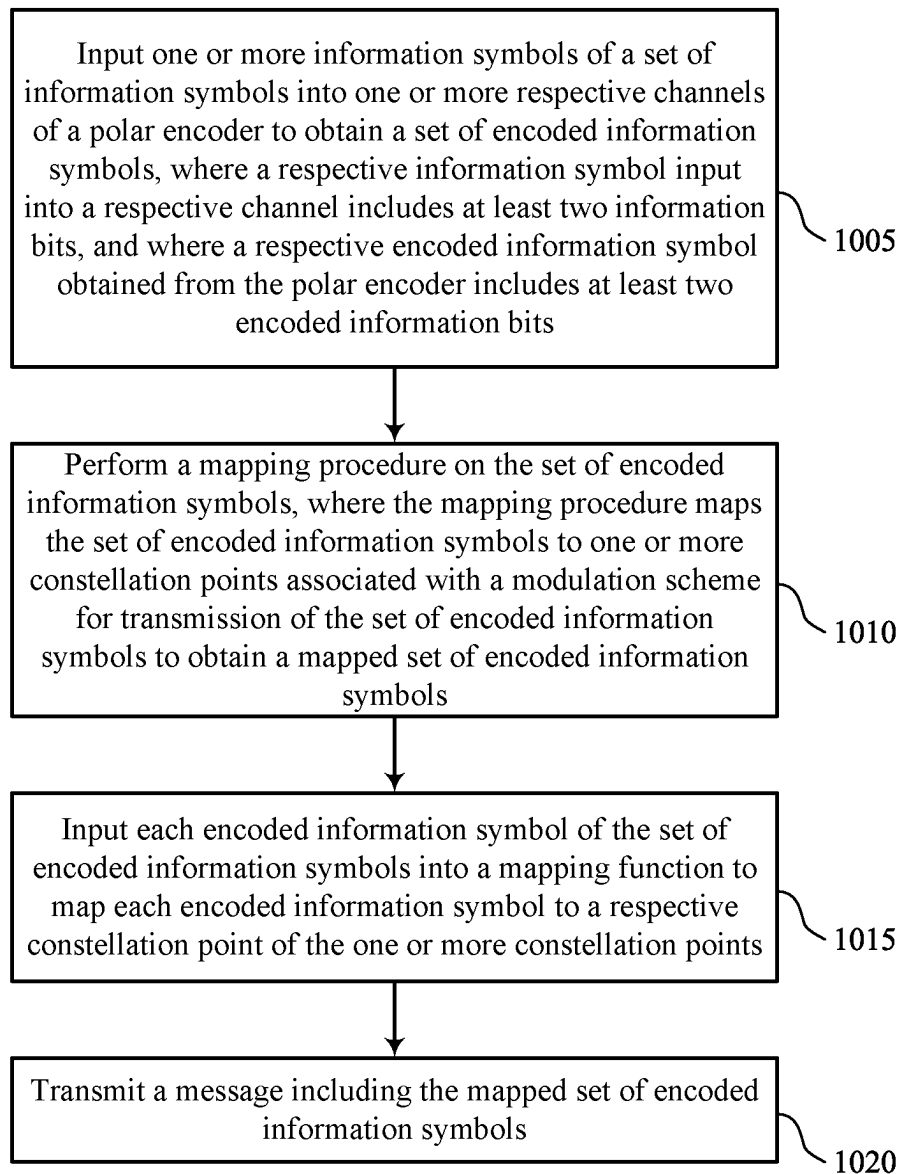
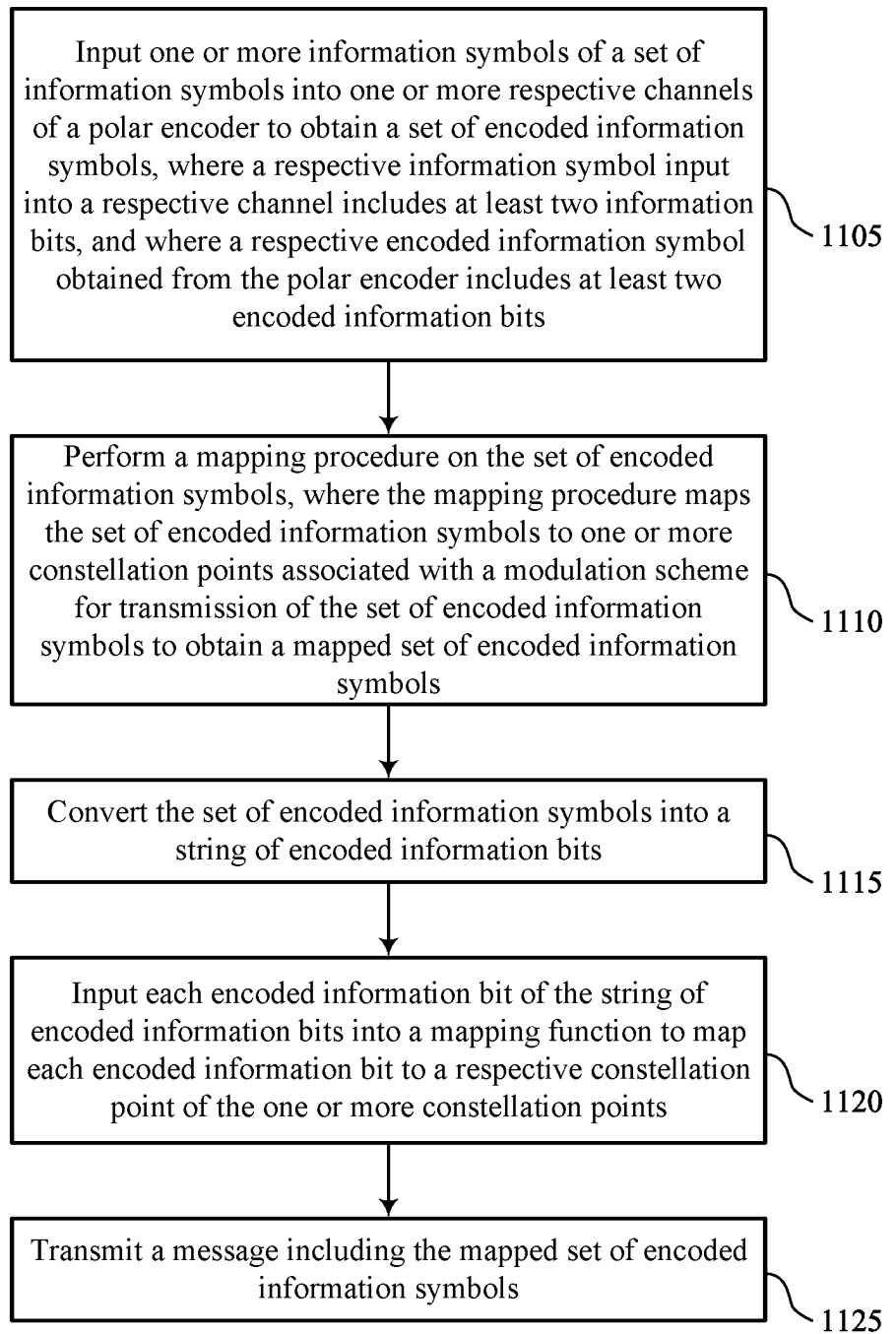


FIG. 9



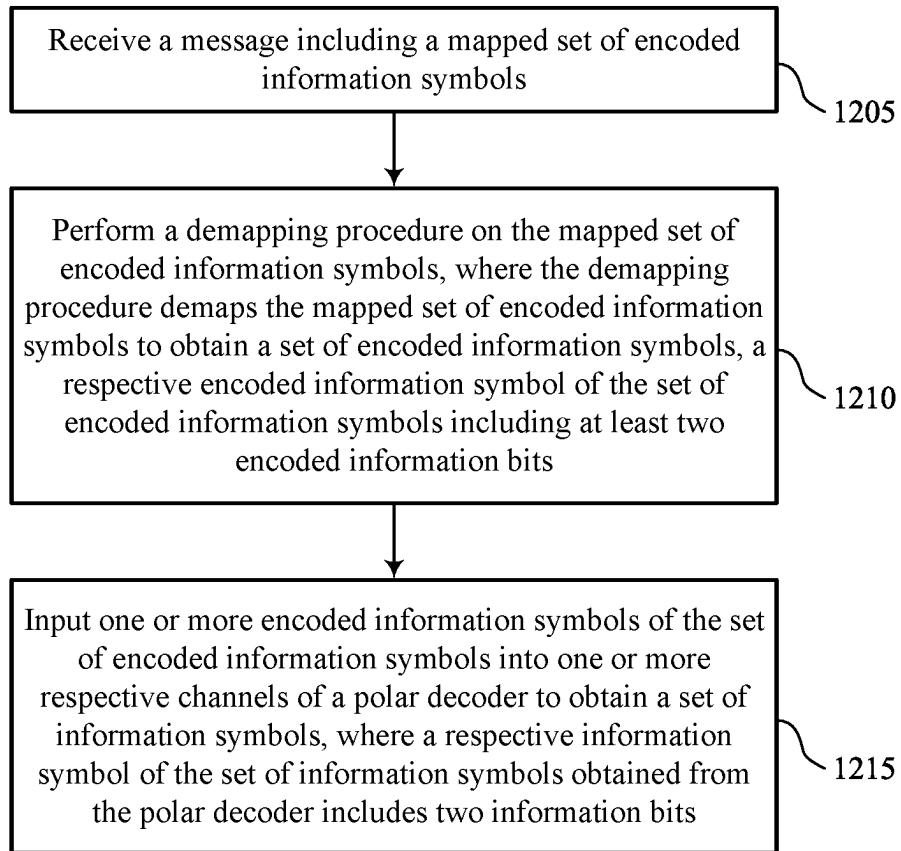
1000

FIG. 10



1100

FIG. 11



1200

FIG. 12

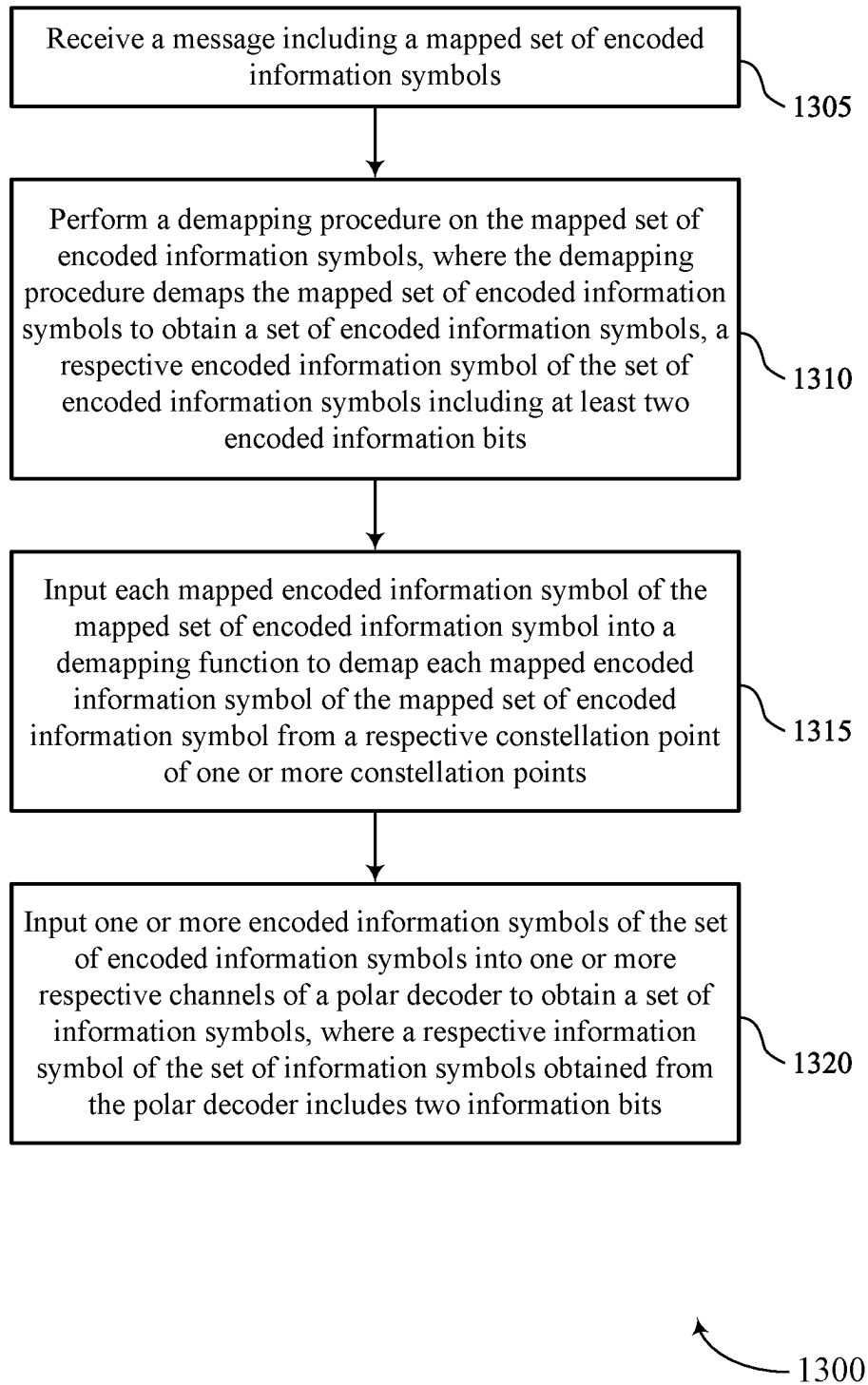
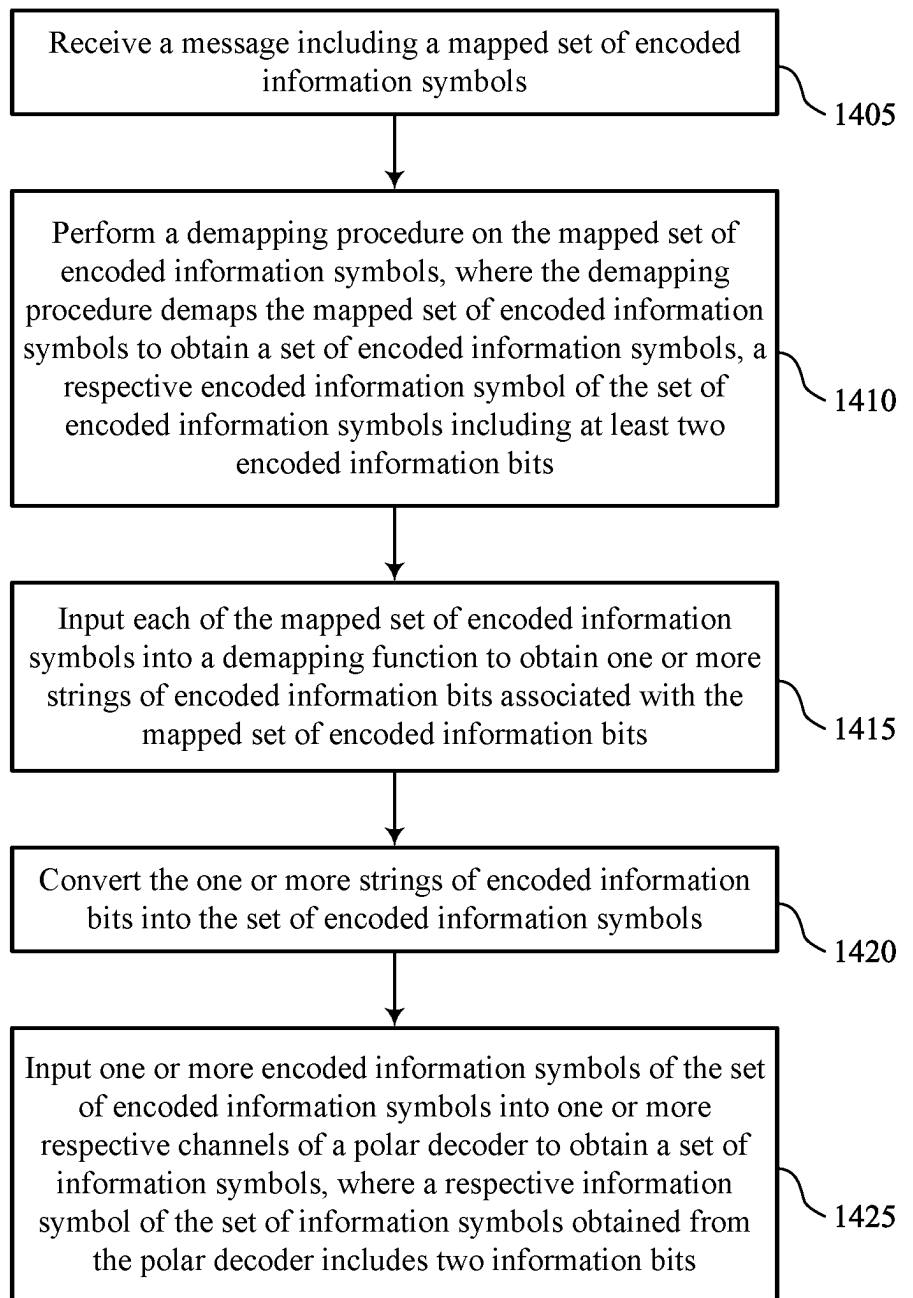


FIG. 13



1400

FIG. 14

MODULATION SCHEMES FOR NONBINARY POLAR CODING

FIELD OF TECHNOLOGY

[0001] The following relates to wireless communications, including modulation schemes for nonbinary polar coding.

BACKGROUND

[0002] Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).

SUMMARY

[0003] The described techniques relate to improved methods, systems, devices, and apparatuses that support modulation schemes for nonbinary polar coding. For example, the described techniques enable a transmitting device to perform a modulation of a set of symbols (e.g., bit pairs) that are encoded using a nonbinary polar encoding scheme. For example, the transmitting device may map each symbol of a set of encoded symbols to a constellation point. Additionally, or alternatively, the transmitting device may convert the encoded symbols into strings of bits. The transmitting device may accordingly perform bit-to-constellation mapping of the resulting bitstring using an a mapping scheme, which may be a new mapping scheme defined for nonbinary polar coding and mapping, as described herein. After transmitting the mapped bitstring, a receiving device may demodulate the received message by performing a demodulation of the set of modulated symbols or bits (e.g., by converting the bitstring into the set of information symbols).

[0004] A method for wireless communications by a wireless device is described. The method may include inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits, performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to

obtain a mapped set of encoded information symbols, and transmitting a message including the mapped set of encoded information symbols.

[0005] A wireless device for wireless communications is described. The wireless device may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the wireless device to input one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits, perform a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols, and transmit a message including the mapped set of encoded information symbols.

[0006] Another wireless device for wireless communications is described. The wireless device may include means for inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits, means for performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols, and means for transmitting a message including the mapped set of encoded information symbols.

[0007] A non-transitory computer-readable medium storing code for wireless communications is described. The code may include instructions executable by one or more processors to input one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits, perform a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols, and transmit a message including the mapped set of encoded information symbols.

[0008] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the mapping procedure may include operations, features, means, or instructions for inputting each encoded information symbol of the set of encoded

information symbols into a mapping function to map each encoded information symbol to a respective constellation point of the one or more constellation points.

[0009] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, the mapping function may be a symbol-level gray labeling function.

[0010] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the mapping procedure may include operations, features, means, or instructions for mapping a first encoded information symbol of the set of encoded information symbols to a set of most significant bits of a first constellation point of the one or more constellation points and mapping a second encoded information symbol of the set of encoded information symbols to a set of least significant bits of the first constellation point.

[0011] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the mapping procedure may include operations, features, means, or instructions for converting the set of encoded information symbols into a string of encoded information bits and inputting each encoded information bit of the string of encoded information bits into a mapping function to map each encoded information bit to a respective constellation point of the one or more constellation points.

[0012] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the mapping procedure may include operations, features, means, or instructions for mapping a first set of multiple bits of the string of encoded information bits to a first constellation point of the one or more constellation points and mapping a second set of multiple bits of the string of encoded information bits to a second constellation point of the one or more constellation points.

[0013] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, one or more reliability levels associated with the mapped set of encoded information symbols may be below a threshold reliability.

[0014] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, the mapping function may be a bit-level gray labeling function.

[0015] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, inputting the one or more information symbols into the one or more respective channels of the polar encoder may include operations, features, means, or instructions for inputting each respective one of the set of information symbols into a respective channel of the polar encoder.

[0016] Some examples of the method, wireless devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for selecting a mapping function associated with the mapping procedure based on a gain associated with the mapping function, an error rate associated with the mapping function, a signal-to-noise ratio associated with a wireless channel, or any combination thereof.

[0017] A method for wireless communications by a wireless device is described. The method may include receiving a message including a mapped set of encoded information symbols, performing a demapping procedure on the mapped

set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits, and inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0018] A wireless device for wireless communications is described. The wireless device may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the wireless device to receive a message including a mapped set of encoded information symbols, perform a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits, and input one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0019] Another wireless device for wireless communications is described. The wireless device may include means for receiving a message including a mapped set of encoded information symbols, means for performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits, and means for inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0020] A non-transitory computer-readable medium storing code for wireless communications is described. The code may include instructions executable by one or more processors to receive a message including a mapped set of encoded information symbols, perform a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits, and input one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0021] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the demapping procedure may include operations, features, means, or instructions for inputting each mapped encoded information symbol of the mapped set of encoded information symbol into a demapping function to demap each mapped encoded information symbol of the mapped set of encoded information symbol from a respective constellation point of one or more constellation points.

[0022] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the demapping procedure may include operations, features, means, or instructions for inputting each of the mapped set of encoded information symbols into a demapping function to obtain one or more strings of encoded information bits associated with the mapped set of encoded information bits and converting the one or more strings of encoded information bits into the set of encoded information symbols.

[0023] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, performing the demapping procedure may include operations, features, means, or instructions for inputting each of the mapped set of encoded information symbols into a symbol-level max-log-map demapper.

[0024] Some examples of the method, wireless devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for obtaining a respective logarithmic probability mass function associated with each of the set of information symbols.

[0025] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, each respective logarithmic probability mass function may be based on a set of multiple probabilities associated with possible values of a respective one of the set of information symbols.

[0026] In some examples of the method, wireless devices, and non-transitory computer-readable medium described herein, inputting the one or more encoded information symbols into the one or more respective channels of the polar decoder may include operations, features, means, or instructions for inputting each respective one of the set of encoded information symbols into a respective channel of the polar decoder.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 shows an example of a wireless communications system that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0028] FIG. 2 shows an example of a wireless communications system that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0029] FIG. 3 shows an example of a block diagram that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0030] FIG. 4 shows an example of a process flow that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0031] FIGS. 5 and 6 show block diagrams of devices that support modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0032] FIG. 7 shows a block diagram of a communications manager that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0033] FIG. 8 shows a diagram of a system including a device that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

[0034] FIGS. 9 through 14 show flowcharts illustrating methods that support modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0035] In some wireless communications systems, a transmitting device may perform encoding and modulation prior to transmitting a string of information bits (e.g., k information bits). For example, the transmitting device may encode N bits (e.g., including the k information bits), and may map each of the N encoded bits to a constellation point defined by a modulation scheme. Such techniques may result in a relatively higher likelihood that a receiving device may successfully receive and decode the k information bits. In some examples, however, the transmitting device may perform nonbinary polar encoding on a set of information symbols, each information symbol including more than one (e.g., two) bits. The transmitting device may therefore generate a set of encoded symbols each including multiple bits (e.g., two bits per symbol). In such examples, the transmitting device may not map the encoded symbols (e.g., bit pairs, set of bits representative of a nonbinary number) in a same manner as encoded single bits (e.g., a single encoded binary bit representative of a binary number).

[0036] Accordingly, in some examples, a transmitting device may perform a modulation of a set of encoded symbols (e.g., bit pairs) that are encoded using a nonbinary polar encoding scheme. For example, the transmitting device may map each symbol of a set of encoded symbols to a constellation point. Additionally, or alternatively, the transmitting device may convert the encoded symbols into strings of bits. For example, the transmitting device may convert a set of four symbols into a string of eight bits. The transmitting device may accordingly perform bit-to-constellation mapping of the resulting bitstring using a mapping scheme. A receiving device may demodulate a received message by performing a demodulation of the set of modulated symbols or bits (e.g., and by converting the bitstring into the set of information symbols).

[0037] Accordingly, in some examples, a transmitting device may perform a modulation of a set of symbols (e.g., bit pairs) that are encoded using a nonbinary polar encoding scheme. For example, the transmitting device may map each symbol of a set of encoded symbols to a constellation point. Additionally, or alternatively, the transmitting device may convert the encoded symbols into strings of bits. The transmitting device may accordingly perform bit-to-constellation mapping of the resulting bitstring according to a mapping scheme.

[0038] Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects of

the disclosure are further illustrated by and described with reference to block diagrams, process flows, apparatus diagrams, system diagrams, and flowcharts that relate to modulation schemes for nonbinary polar coding.

[0039] FIG. 1 shows an example of a wireless communications system 100 that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The wireless communications system 100 may include one or more devices, such as one or more network devices (e.g., network entities 105), one or more UEs 115, and a core network 130. In some examples, the wireless communications system 100 may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0040] The network entities 105 may be dispersed throughout a geographic area to form the wireless communications system 100 and may include devices in different forms or having different capabilities. In various examples, a network entity 105 may be referred to as a network element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities 105 and UEs 115 may wirelessly communicate via communication link(s) 125 (e.g., a radio frequency (RF) access link). For example, a network entity 105 may support a coverage area 110 (e.g., a geographic coverage area) over which the UEs 115 and the network entity 105 may establish the communication link(s) 125. The coverage area 110 may be an example of a geographic area over which a network entity 105 and a UE 115 may support the communication of signals according to one or more radio access technologies (RATs).

[0041] The UEs 115 may be dispersed throughout a coverage area 110 of the wireless communications system 100, and each UE 115 may be stationary, or mobile, or both at different times. The UEs 115 may be devices in different forms or having different capabilities. Some example UEs 115 are illustrated in FIG. 1. The UEs 115 described herein may be capable of supporting communications with various types of devices in the wireless communications system 100 (e.g., other wireless communication devices, including UEs 115 or network entities 105), as shown in FIG. 1.

[0042] As described herein, a node of the wireless communications system 100, which may be referred to as a network node, or a wireless node, may be a network entity 105 (e.g., any network entity described herein), a UE 115 (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE 115. As another example, a node may be a network entity 105. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE 115, the second node may be a network entity 105, and the third node may be a UE 115. In another aspect of this example, the first node may be a UE 115, the second node may be a network entity 105, and the third node may be a network entity 105. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE 115, network entity 105, apparatus, device, computing system, or

the like may include disclosure of the UE 115, network entity 105, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE 115 is configured to receive information from a network entity 105 also discloses that a first node is configured to receive information from a second node.

[0043] In some examples, network entities 105 may communicate with a core network 130, or with one another, or both. For example, network entities 105 may communicate with the core network 130 via backhaul communication link(s) 120 (e.g., in accordance with an S1, N2, N3, or other interface protocol). In some examples, network entities 105 may communicate with one another via backhaul communication link(s) 120 (e.g., in accordance with an X2, Xn, or other interface protocol) either directly (e.g., directly between network entities 105) or indirectly (e.g., via the core network 130). In some examples, network entities 105 may communicate with one another via a midhaul communication link 162 (e.g., in accordance with a midhaul interface protocol) or a fronthaul communication link 168 (e.g., in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication link(s) 120, midhaul communication links 162, or fronthaul communication links 168 may be or include one or more wired links (e.g., an electrical link, an optical fiber link) or one or more wireless links (e.g., a radio link, a wireless optical link), among other examples or various combinations thereof. A UE 115 may communicate with the core network 130 via a communication link 155.

[0044] One or more of the network entities 105 or network equipment described herein may include or may be referred to as a base station 140 (e.g., a base transceiver station, a radio base station, an NR base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity 105 (e.g., a base station 140) may be implemented in an aggregated (e.g., monolithic, standalone) base station architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within one network entity (e.g., a network entity 105 or a single RAN node, such as a base station 140).

[0045] In some examples, a network entity 105 may be implemented in a disaggregated architecture (e.g., a disaggregated base station architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among multiple network entities (e.g., network entities 105), such as an integrated access and backhaul (IAB) network, an open RAN (O-RAN) (e.g., a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (e.g., a cloud RAN (C-RAN)). For example, a network entity 105 may include one or more of a central unit (CU), such as a CU 160, a distributed unit (DU), such as a DU 165, a radio unit (RU), such as an RU 170, a RAN Intelligent Controller (RIC), such as an RIC 175 (e.g., a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) system, such as an SMO system 180, or any combination thereof. An RU 170 may also be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or

more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (e.g., separate physical locations). In some examples, one or more of the network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (e.g., a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0046] The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending on which functions (e.g., network layer functions, protocol layer functions, baseband functions, RF functions, or any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some examples, the CU **160** may host upper protocol layer (e.g., layer 3 (L3), layer 2 (L2)) functionality and signaling (e.g., Radio Resource Control (RRC), service data adaptation protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** (e.g., one or more CUs) may be connected to a DU **165** (e.g., one or more DUs) or an RU **170** (e.g., one or more RUs), or some combination thereof, and the DUs **165**, RUs **170**, or both may host lower protocol layers, such as layer 1 (L1) (e.g., physical (PHY) layer) or L2 (e.g., radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the protocol stack. The DU **165** may support one or multiple different cells (e.g., via one or multiple different RUs, such as an RU **170**). In some cases, a functional split between a CU **160** and a DU **165** or between a DU **165** and an RU **170** may be within a protocol layer (e.g., some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to a DU **165** via a midhaul communication link **162** (e.g., F1, F1-c, F1-u), and a DU **165** may be connected to an RU **170** via a fronthaul communication link **168** (e.g., open fronthaul (FH) interface). In some examples, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (e.g., a channel) between layers of a protocol stack supported by respective network entities (e.g., one or more of the network entities **105**) that are in communication via such communication links.

[0047] In some wireless communications systems (e.g., the wireless communications system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (e.g., to a core network **130**). In some cases, in an IAB network, one or more of the network entities **105** (e.g., network entities **105** or IAB node(s) **104**) may be partially controlled by each other. The IAB node(s) **104** may be referred to as a donor

entity or an IAB donor. A DU **165** or an RU **170** may be partially controlled by a CU **160** associated with a network entity **105** or base station **140** (such as a donor network entity or a donor base station). The one or more donor entities (e.g., IAB donors) may be in communication with one or more additional devices (e.g., IAB node(s) **104**) via supported access and backhaul links (e.g., backhaul communication link(s) **120**). IAB node(s) **104** may include an IAB mobile termination (IAB-MT) controlled (e.g., scheduled) by one or more DUs (e.g., DUs **165**) of a coupled IAB donor. An IAB-MT may be equipped with an independent set of antennas for relay of communications with UEs **115** or may share the same antennas (e.g., of an RU **170**) of IAB node(s) **104** used for access via the DU **165** of the IAB node(s) **104** (e.g., referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB node(s) **104** may include one or more DUs (e.g., DUs **165**) that support communication links with additional entities (e.g., IAB node(s) **104**, UEs **115**) within the relay chain or configuration of the access network (e.g., downstream). In such cases, one or more components of the disaggregated RAN architecture (e.g., the IAB node(s) **104** or components of the IAB node(s) **104**) may be configured to operate according to the techniques described herein.

[0048] In the case of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support test as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (e.g., a base station **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (e.g., components such as an IAB node, a DU **165**, a CU **160**, an RU **170**, an RIC **175**, an SMO system **180**).

[0049] A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, vehicles, or meters, among other examples.

[0050] The UEs **115** described herein may be able to communicate with various types of devices, such as UEs **115** that may sometimes operate as relays, as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. 1.

[0051] The UEs **115** and the network entities **105** may wirelessly communicate with one another via the communication link(s) **125** (e.g., one or more access links) using resources associated with one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined PHY layer structure for supporting the communication link(s) **125**. For example, a carrier used for the communication link(s) **125** may include a portion of an RF

spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more PHY layer channels for a given RAT (e.g., LTE, LTE-A, LTE-A Pro, NR). Each PHY layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system 100 may support communication with a UE 115 using carrier aggregation or multi-carrier operation. A UE 115 may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity 105 and other devices may refer to communication between the devices and any portion (e.g., entity, sub-entity) of a network entity 105. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity 105, may refer to any portion of a network entity 105 (e.g., a base station 140, a CU 160, a DU 165, a RU 170) of a RAN communicating with another device (e.g., directly or via one or more other network entities, such as one or more of the network entities 105).

[0052] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, in which case the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (e.g., in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (e.g., a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE 115.

[0053] The time intervals for the network entities 105 or the UEs 115 may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of $T_s = 1/(\Delta f_{max} N_f)$ seconds, for which Δf_{max} may represent a supported subcarrier spacing, and N_f may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).

[0054] Each frame may include multiple consecutively-numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (e.g., depending on the length of the cyclic prefix prepended

to each symbol period). In some wireless communications systems, such as the wireless communications system 100, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (e.g., N_f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

[0055] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system 100 and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system 100 may be dynamically selected (e.g., in bursts of shortened TTIs (STTIs)).

[0056] Physical channels may be multiplexed for communication using a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs 115. For example, one or more of the UEs 115 may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to UEs 115 (e.g., one or more UEs) or may include UE-specific search space sets for sending control information to a UE 115 (e.g., a specific UE).

[0057] In some examples, a network entity 105 (e.g., a base station 140, an RU 170) may be movable and therefore provide communication coverage for a moving coverage area, such as the coverage area 110. In some examples, coverage areas 110 (e.g., different coverage areas) associated with different technologies may overlap, but the coverage areas 110 (e.g., different coverage areas) may be supported by the same network entity (e.g., a network entity 105). In some other examples, overlapping coverage areas, such as a coverage area 110, associated with different technologies may be supported by different network entities (e.g., the network entities 105). The wireless communications system 100 may include, for example, a heterogeneous network in which different types of the network entities 105 support communications for coverage areas 110 (e.g., different coverage areas) using the same or different RATs.

[0058] The wireless communications system 100 may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system 100 may be configured to support ultra-reliable low-latency commu-

nications (URLLC). The UEs 115 may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

[0059] In some examples, a UE 115 may be configured to support communicating directly with other UEs (e.g., one or more of the UEs 115) via a device-to-device (D2D) communication link, such as a D2D communication link 135 (e.g., in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some examples, one or more UEs 115 of a group that are performing D2D communications may be within the coverage area 110 of a network entity 105 (e.g., a base station 140, an RU 170), which may support aspects of such D2D communications being configured by (e.g., scheduled by) the network entity 105. In some examples, one or more UEs 115 of such a group may be outside the coverage area 110 of a network entity 105 or may be otherwise unable to or not configured to receive transmissions from a network entity 105. In some examples, groups of the UEs 115 communicating via D2D communications may support a one-to-many (1:M) system in which each UE 115 transmits to one or more of the UEs 115 in the group. In some examples, a network entity 105 may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs 115 without an involvement of a network entity 105.

[0060] The core network 130 may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network 130 may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs 115 served by the network entities 105 (e.g., base stations 140) associated with the core network 130. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services 150 for one or more network operators. The IP services 150 may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0061] The wireless communications system 100 may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the

waves may penetrate structures sufficiently for a macro cell to provide service to the UEs 115 located indoors. Communications using UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than one hundred kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0062] The wireless communications system 100 may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system 100 may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) RAT, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities 105 and the UEs 115 may employ carrier sensing for collision detection and avoidance. In some examples, operations using unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (e.g., LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0063] A network entity 105 (e.g., a base station 140, an RU 170) or a UE 115 may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity 105 or a UE 115 may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity 105 may be located at diverse geographic locations. A network entity 105 may include an antenna array with a set of rows and columns of antenna ports that the network entity 105 may use to support beamforming of communications with a UE 115. Likewise, a UE 115 may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0064] Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a network entity 105, a UE 115) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect

to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0065] To transmit information, a transmitting device (e.g., a UE **115**, a network entity **105**) may encode a set of information bits via an encoder. An encoder may have a finite input bus width or input bits of a single input vector may become available at different times. For example, an input vector may include information bits, check bits, frozen bits, etc., where some or all information bits may be generated by or received from different sources. For example, bits that are not allocated as information bits may be assigned as parity bits or frozen bits. Parity bits may be used in parity check (PC) polar coding techniques and frozen bits may be bits of a given value (0, 1, etc.) known to the both the encoder and decoder (e.g., the encoder encoding information bits at a transmitter and the decoder decoding the codeword received at a receiver). Construction of an input vector may depend on information bit location/distribution amongst the channel instances of the encoder. In some cases, different input vectors containing the same information bits (e.g., input vectors with a different allocation of information bits) may be received with different success rates (e.g., encoded according to different reliabilities). For example, an input vector with information bits distributed such that the information bits are loaded to channel instances associated with high reliability metrics may be decoded at an increased success rate. The input vector may be, for example, a physical channel message (e.g., control channel message) or a data or shared channel message (e.g., a data packet).

[0066] System performance of transmission of information bits in low-latency environments may be determined by factors such as overhead, coding gain, transmission pipelining, and decoding delay. Some processing techniques may emphasize improving transmission pipelining and decoding delay at the expense of higher overhead and lower coding gain. Generally, use of a larger code length (e.g., larger codeword) provides higher coding gain. However, a larger code length result in a larger decoding delay and overall system latency. In contrast, smaller code length reduces latency or decoding delay, but may result in an increase in overhead or lower coding gain.

[0067] In some examples, the transmitting device (e.g., a network entity **105**, a UE **115**) may perform a modulation of a set of encoded symbols (e.g., bit pairs) that are encoded using a nonbinary polar encoding scheme. For example, the transmitting device may map each symbol of a set of encoded symbols to a constellation point. Additionally, or alternatively, the transmitting device may convert the encoded symbols into strings of bits. For example, the transmitting device may convert a set of four symbols into a string of eight bits (or more bits depending on the quantity of bits per symbol). The transmitting device may accordingly perform bit-to-constellation mapping of the resulting bitstring using a mapping scheme. A receiving device (e.g., a network entity **105**, a UE **115**) may demodulate a received message by performing a demodulation of the set of modulated symbols or bits (e.g., and by converting the bitstring into the set of information symbols).

[0068] FIG. 2 shows an example of a wireless communications system **200** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The wireless communications system **200** may implement or may be implemented by aspects of the wireless communications system **100**. For

example, the wireless communications system **200** may be implemented by one or more wireless devices **205**, which may be examples of UEs **115** or network entities **105** as described with reference to FIG. 1.

[0069] In some examples, a wireless device **205-a** may transmit a message **215** to a wireless device **205-b** via a channel **210**. The message **215** may include, for example, a control message, a data message, or another type of message. The message **215** may indicate one or more bits of information to the wireless device **205-b** via modulation. That is, the wireless device **205-a** may map one or more bits to constellation points according to a modulation scheme. Each constellation point may be indicative of an amplitude (e.g., power) of a signal and a phase shift of the signal (e.g., relative to a reference signal), where the message **215** is conveyed via the signal.

[0070] Each constellation point may be a point in a constellation diagram, where the amplitude and phase shift associated with each constellation point is defined according to a distance of the constellation point from an origin of a set of axes (e.g., a point at which a horizontal axis and a vertical axis intersect) and an angle of the constellation point measured (e.g., counter-clockwise) from the horizontal axis, respectively. For example, in an 8 phase shift keying (PSK) modulation scheme, a point that is a distance 1 from the origin and a distance of 90° from the horizontal axis may indicate three bits (e.g., 010). The wireless device **205-a** may accordingly convey the bits 010 to the wireless device **205-b** by modulating the carrier signal to have an amplitude of 1 and a phase shift of 90°. To convey a set of bits, the wireless device **205-a** may perform a bit-to-constellation mapping (e.g., as a function of bit groups, such as octets for 256 quadrature amplitude modulation (QAM) modulation).

[0071] Each constellation point may be indicative of a quantity of bits. For example, for 16QAM, the wireless device **205-a** may map the bits to 16 different constellation points in a 16QAM constellation diagram, and each constellation point may be indicative of four bits. Other modulation schemes (e.g., 8QAM, 256QAM, quadrature phase shift keying (QPSK), binary phase shift keying (BPSK)) may indicate other quantities of bits per constellation point. In some examples, a modulation scheme with relatively more bits per constellation point (e.g., a higher-order modulation scheme) may result in reduced latency as compared to a modulation scheme with relatively fewer bits per constellation point as a result of transmitting relatively more bits at a time. However, such higher-order modulation schemes may have a relatively lower likelihood of a message **215** being correctly received and decoded by the wireless device **205-b** (e.g., due to noise via the channel **210**, interference, and a relatively smaller distance between constellation points on a respective constellation diagram of the higher-order modulation scheme).

[0072] In some examples, to increase a likelihood that the wireless device **205-b** may successfully receive and decode a message **215**, the wireless device **205-a** may encode the message **215** using a polar encoding scheme. That is, the wireless device **205-a** may input a set of bits (e.g., a set of information bits and a set of frozen bits with a set value of 0 or 1) into a corresponding set of channels (e.g., channels u_0 through u_7). The channels may output an encoded set of bits that include information about the input set of bits (e.g., via one or more exclusionary or (XOR) operations performed on the set of channels). The encoded set of bits may

include relatively more information related to some channels (e.g., a first subset of channels) and relatively less information related to some other channels (e.g., a second subset of channels). Accordingly, to increase a likelihood that the wireless device **205-b** may successfully receive the set of information bits, the wireless device **205-a** may input the information bits (e.g., and one or more bits related to performing integrity checks and the like, such as parity bits or a cyclic redundancy check (CRC)) into the first subset of channels and the set of frozen bits into the second subset of channels.

[0073] The wireless device **205-a** may perform a modulation on the encoded set of bits by mapping each of the encoded set of bits to a constellation point. The wireless device **205-a** may determine a constellation point to map each of the encoded set of bits to using a bit-to-constellation mapping scheme, such as a Gray labeling scheme. The bit-to-constellation mapping scheme may be, for example, a function which defines a constellation point for each of the encoded set of bits.

[0074] In some aspects, the wireless device **205-a** may perform a nonbinary polar coding procedure. For example, for a 4-ary polar code, the wireless device **205-a** may encode a set of information symbols including a two information bit payload (e.g., and a set of frozen symbols with bits set to 00). That is, the wireless device **205-a** may input each information symbol the set of information symbols and the set of frozen symbols (e.g., the symbols u_0 through u_7 , as illustrated with reference to FIG. 2) into respective channels of a nonbinary encoder **220** (e.g., channels $W^{(\text{---})}$ through $W^{(++)}$), as illustrated with reference to FIG. 2). For one or more other nonbinary polar codes, the wireless device **205-a** may input sets of information and frozen symbols including different quantities of information bits (e.g., greater than two bits per symbol) into channels of a nonbinary encoder **220**.

[0075] The encoder may encode the set of symbols u with $N=2^m$, K code defined over a finite field $GF(4)$ where K may represent the quantity of information symbols (e.g., information symbols i_0 through i_{K-1}), N may represent the quantity of polar coded bits or the polar code length (e.g., a power of 2 defined using a 2 by 2 kernel G), and m may represent an integer such that $2^m=N$. The elements of the finite field $GF(4)$ (e.g., or $GF(2^2)$) may be represented by $\{0, 1, \alpha, \alpha^2=\alpha+1\}$, or by $\{00, 01, 10, 11\}$ of two-dimensional vector space over $GF(2)$. That is, a binary representation of coded symbols given by $\phi: GF(4) \rightarrow \{0,1\}^2$ may be as follows: $0 \rightarrow 00, 1 \rightarrow 01, 2 \rightarrow 10, 3 \rightarrow 11$.

[0076] To encode the set of symbols using the nonbinary encoder **220**, the wireless device **205-a** may perform operations across the set of channels W . For example, the wireless device **205-a** may multiple the symbol u_i by a general nonbinary kernel $G=$

$$\begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}^{\otimes m}$$

(e.g., with a recursive structure), where γ is a non-zero element of $GF(4)$ (e.g., $\gamma \in \{1, \alpha, \alpha^2\}$, $\gamma \in \{01, 10, 11\}$), and where m is defined as an integer such that $2^m=N$. The multiplication operation of a symbol u_i with the kernel G may generate a matrix c_i , defined as

$$c_i = u_i \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}^{\otimes m},$$

that satisfies $u_i, c_i \in GF(4)$. For $GF(4)$ with elements $\{0, 1, \alpha, \alpha^2=\alpha+1\}$, the multiplication operation of a bit x with γ may result in $0 \cdot x=0$, $1 \cdot x=x$, or $\alpha^3=1$. The multiplication operations are illustrated with reference to FIG. 2 by the symbol γ .

[0077] The wireless device **205-b** may add the resulting matrix c_i to a symbol input into a different channel W . The addition operations may be performed coordinate-wise (e.g., bit-wise). The addition operations are illustrated with reference to FIG. 2 by the symbol $+$.

[0078] In some examples, if γ is not equal to 1 (e.g., 01, as represented in binary), the matrix

$$\begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}^{\otimes m}$$

may polarize 2^m copies of a given channel W into subchannels $W^{(i)}$. A subchannel $W^{(i)}$ may be almost noisy (e.g., with a lower channel quality and a lower likelihood of being decoded correctly) if an operation $I(W^{(i)})$ approaches 0, or almost noiseless (e.g., with a higher channel quality and a higher likelihood of being decoded correctly) if the operation $I(W^{(i)})$ approaches 2. That is, a higher value of the operation $I(W^{(i)})$ may correspond to a higher capacity (e.g., a higher likelihood of being decoded correctly). In some examples, if γ may be equal to 1, the operation $I(W^{(i)})$ on some subchannels $W^{(i)}$ may approach 1. The wireless device **205-a** may map the set of information symbols i_0 through i_{K-1} to subchannels corresponding to the K highest values of the operation $I(W^{(i)})$.

[0079] The nonbinary encoder **220** may output a set of encoded symbols (e.g., a codeword of encoded symbols W_0 through W_7). Each encoded symbol may have multiple encoded bits (e.g., two encoded bits for a 4-ary polar code). The wireless device **205-a** may perform a modulation on the set of encoded symbols by mapping the encoded symbols to constellation points. That is, the wireless device **205-b** may perform a codeword-to-constellation mapping (e.g., to increase or maximize channel performance, such as by increasing a signal-to-noise ratio (SNR) or increasing a likelihood that the wireless device **205-b** may decode the message **215** correctly).

[0080] In some aspects, the wireless device **205-a** may perform a direct symbol-to-constellation mapping by defining a mapping as a function of $GF(4)$ symbols. That is, the wireless device **205-a** may map each symbol to a respective constellation point. Additionally, or alternatively, the wireless device **205-a** may perform an indirect codeword-to-constellation mapping by converting tuples of $GF(4)$ symbols to bit strings, and using a bit-to constellation mapping function (e.g., Gray labeling or another function defining a constellation point mapped to each bit). Accordingly (e.g., for a 16QAM modulation scheme), the wireless device **205-a** may map both bits of an encoded symbol c_0 to a same 4-pulse amplitude modulation (PAM) symbol (e.g., constellation point) with two bits per constellation point or may map each bit of c_0 to a different 4-PAM symbol (e.g.,

constellation point) symbol-level max-log-map demappers. Such mapping techniques are described in further detail with reference to FIG. 3.

[0081] The wireless device **205-b** may receive the message **215** (e.g., as a set of modulated and encoded symbols) and perform a demapping (e.g., demodulation) procedure to generate a set of encoded symbols. For example, the wireless device **205-b** may perform a symbol-level demodulation procedure using a symbol-level max-log-map demapper, or a bit-level max-log-map demapper (e.g., to independently demodulate bits b_0 , b_1 of a symbol c_0). The bit-level max-log-map demapper may be represented according to Equations 1 and 2 below, where Equation 1 represents a demapping of bit b_0 and Equation 2 represents a demapping of bit b_1 .

$$\log\left(\frac{p(b_0 = 1 | y)}{p(b_0 = 0 | y)}\right) = \frac{(y - \mu_{0,0}^*(y))^2 - (y - \mu_{1,0}^*(y))^2}{2\sigma^2} \quad (1)$$

$$\log\left(\frac{p(b_1 = 1 | y)}{p(b_1 = 0 | y)}\right) = \frac{(y - \mu_{0,1}^*(y))^2 - (y - \mu_{1,1}^*(y))^2}{2\sigma^2} \quad (2)$$

[0082] As described with reference to Equations 1 and 2, $p(b_0=i|y)$ and $p(b_1=i|y)$ may be probabilities that bit b_0 and bit b_1 are equal to a value i , respectively, $\mu_{i,0}^*(y)$ and $\mu_{i,1}^*(y)$ may be the closest constellation points to a symbol y with $b_0=i$ and $b_1=i$, respectively, and σ may be a standard distribution of normally-distributed noise associated with a wireless channel between the encoding device and the decoding device.

[0083] If both bits of a symbol are mapped to a same constellation point, the wireless device **205-b** may use a symbol-level max-log-map demapper, represented according to Equation 3 below.

$$\log\left(\frac{p(c_0 = i | y)}{p(c_0 = 0 | y)}\right) = \frac{(y - \mu_{0,0}^*(y))^2 - (y - \mu_{1,0}^*(y))^2}{2\sigma^2} \quad (3)$$

[0084] As described with reference to Equation 3, $p(c_0=i|y)$ may be a probability that the symbol c_0 is equal to a value i , $\mu_{i,0}^*(y)$ may be the closest constellation points corresponding to $c_0=i$, respectively, and σ may be the standard distribution of normally-distributed noise associated with the wireless channel between the encoding device and the decoding device. In some examples (e.g., as the wireless device **205-b** may demodulate a whole symbol, rather than individual bits), mappings from GF(4) to 4-PAM symbols (e.g., constellation points) may each have a same performance (e.g., a same change of being decoded correctly).

[0085] In some examples, for a higher-order QAM scheme (e.g., 256 QAM with 16-PAM modulation symbols or constellation points and four bits per constellation point), the wireless device **205-a** may map two bits of a first coded GF(4) symbol c_0 to two most significant bits (MSBs) of a modulation symbol (e.g., constellation point) x , and may map two bits of a second coded GF(4) symbol c_1 to two least significant bits (LSBs) of x (e.g., symbol-independent coded modulation (SICM)). Such mapping techniques are described in further detail with reference to FIG. 3. In such examples, to demap the message **215** with a symbol-level

max-log-map demapper, the wireless device **205-b** may use Equations 4 and 5 below to demap the symbol c_0 and the symbol c_1 , respectively.

$$\log\left(\frac{p(c_0 = i | y)}{p(c_0 = 0 | y)}\right) = \frac{(y - \mu_{0,0}^*(y))^2 - (y - \mu_{i,0}^*(y))^2}{2\sigma^2} \quad (4)$$

$$\log\left(\frac{p(c_1 = i | y)}{p(c_1 = 0 | y)}\right) = \frac{(y - \mu_{0,1}^*(y))^2 - (y - \mu_{i,1}^*(y))^2}{2\sigma^2} \quad (5)$$

[0086] As described with reference to Equations 4 and 5, $\mu_{i,0}^*(y)$ and $\mu_{i,1}^*(y)$ may represent closest constellation points to a received symbol y that have $\phi(i)$ in MSBs and LSBs, respectively, where $\phi(i)$ is a mapping from elements of GF(4) to bit pairs (e.g., such that $\phi(0)=00$, $\phi(1)=01$, $\phi(2)=10$, $\phi(3)=11$). In some examples, for arbitrary mapping of bits ($\phi(c_0)$, $\phi(c_1)$) (e.g., if the bits of c_0 and c_1 are not mapped to the MSBs and LSBs of x), the wireless device **205-b** may use a different subset of constellation points corresponding to fixed bit values used to compute $\mu_{i,0}^*(y)$ and $\mu_{i,1}^*(y)$.

[0087] The wireless device **205-b** may perform a decoding procedure (e.g., using a nonbinary decoder) to generate a set of symbols y with multiple (e.g., two) bits per symbol, and may determine a value for each bit of each symbol y . The set of symbols y may include information symbols (e.g., with two or more information bits) and frozen symbols.

[0088] To decode the message **215**, the wireless device **205-b** may generate a statistic $p(c|y)$ representing the probability that each received symbol y is equal to a value c . For a nonbinary alphabet (e.g., nonbinary encoding and decoding with multiple bits per symbol), the wireless device **205-b** may generate a logarithmic probability mass function (LMPF) (e.g., rather than a single log likelihood ratio (LLR) value) representing the probability. For example, the wireless device **205-b** may generate, for each symbol y , a LMPF s , where each entry in the LMPF is a piecewise linear function of the received symbols y , defined according to Equation 6 below.

$$s = \quad (6)$$

$$\left[\log\left(\frac{p(c=0|y)}{p(c=0|y)}\right) \log\left(\frac{p(c=1|y)}{p(c=0|y)}\right) \log\left(\frac{p(c=2|y)}{p(c=0|y)}\right) \log\left(\frac{p(c=3|y)}{p(c=0|y)}\right) \right]$$

[0089] As described with reference to Equation 6, $p(c=0|y)$ may represent a probability that a symbol c is equal to 0 (e.g., with bits 00), $p(c=1|y)$ may represent a probability that a symbol c is equal to 1 (e.g., with bits 01), $p(c=2|y)$ may represent a probability that a symbol c is equal to 2 (e.g., with bits 10), and $p(c=3|y)$ may represent a probability that a symbol c is equal to 3 (e.g., with bits 11). The term

$$\log\left(\frac{p(c=0|y)}{p(c=0|y)}\right)$$

may be equal to 0; thus, wireless device **205-b** may use the values of

$$\log\left(\frac{p(c=i|y)}{p(c=0|y)}\right),$$

where $i \neq 0$, to determine the LMPF. The functions

$$\log\left(\frac{p(c=i|y)}{p(c=0|y)}\right)$$

may be the symbol-level max-log-map demappers represented with reference to Equations 3, 4, and 5, or one or more other symbol-level max-log-map demappers.

[0090] FIG. 3 shows an example of a block diagram **300** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The block diagram **300** may implement or may be implemented by aspects of the wireless communications system **100** or the wireless communications system **200**. For example, the block diagram **300** may be implemented by one or more wireless devices, which may be examples of UEs **115** or network entities **105** as described with reference to FIG. 1.

[0091] In some examples, a wireless device may perform a mapping procedure **310** on a set of encoded symbols (e.g., symbols encoded using a nonbinary polar encoder), including a symbol **305-a** (e.g., c_0) with bits (b_0, b_1) and a symbol **305-b** (e.g., c_1) with bits (b_2, b_3) . The set of encoded symbols may include one or more additional symbols **305**, and each symbol **305** may include one or more additional bits b .

[0092] The wireless device may perform the mapping procedure **310** to map each of the symbols **305** to a constellation point x . In some examples, the wireless device may perform symbol-to-constellation mapping **315-a** to map each symbol **305** to a respective constellation point x . For example, for a 256QAM modulation scheme (e.g., with 16-PAM symbols), the wireless device may map the bits of the symbol **305-a** to two or more MSBs of x and the bits of the symbol **305-a** to two or more LSBs of x . For other modulation schemes (e.g., for M^2 -QAM with even $\log_2 M$), the wireless device may map a quantity

$$\frac{\log_2 M}{2}$$

of symbols **305** to a real component of a complex constellation point x (e.g., a modulation symbol x_0) and a quantity

$$\frac{\log_2 M}{2}$$

of symbols **305** to an imaginary component of the complex constellation point x (e.g., a modulation symbol x_1) such that $\text{Re}(x)=x_0$ and $\text{Im}(x)=x_1$.

[0093] In some examples, the symbol-to-constellation mapping **315-a** may use a mapping function to map each symbol **305** to a constellation point. For example, for

256QAM (e.g., 16=PAM), a symbol-to-constellation mapping function may be defined as $(c_0, c_1) \rightarrow (3-2c_0)+4(3-2c_1)$. In general, to map a set of coded symbols **305** (e.g., a quantity of $2i+1$ coded symbols **305** ($c_0, \dots, c_{2i}$)) to $(2i+1)$ -bit modulation symbols (e.g., constellation points), the wireless device may map symbols $(c_0, \dots, c_{i-1})$ and a first bit of symbol c_{2i} to a first modulation symbol (e.g., constellation point) x_0 and a second bit of symbol c_{2i} and symbols $(c_i, \dots, c_{2i-1})$ to a second modulation symbol (e.g., constellation point) x_1 . For $2^{2(2i+1)}$ -QAM symbols, the symbols x_0 and x_1 may be $(2i+1)$ -bit real and imaginary symbols (e.g., I and Q symbols), respectively. That is, the wireless device may map symbols $(c_0, \dots, c_{2i})$ to a complex-valued symbol x such that $\text{Re}(x)=x_0$ and $\text{Im}(x)=x_1$.

[0094] The wireless device may use one or more other symbol-to-constellation mapping functions, such as a symbol-level gray mapping function. In a symbol-level gray mapping, consecutive constellation points (e.g., modulation symbols) may differ in one GF(4) symbol. Gray labeling for SCIM may result in close symbol-level capacities such that a reliability associated with demodulating the symbol **305-a** may be relatively close to a reliability associated with demodulating the symbol **305-b**. For $2^{2(2i+1)}$ -QAM (e.g., 64-QAM), the symbol-to-constellation mapping may be defined on a whole symbol level (e.g., in terms of a whole complex-valued symbol x) as

$$(c_0, c_1) \rightarrow (-1)^{\frac{c_0}{2}} (4c_1 + (-1)^{c_0+c_1} + 2).$$

[0095] Additionally, or alternatively, the wireless device may perform bit-to-constellation mapping **315-b**. For example, the wireless device may convert the symbols c_0 and c_1 to a string of bits (e.g., $(\phi(c_0), \phi(c_1))=(b_0, b_1, b_2, b_3)$) and may perform a mapping of each bit to a corresponding symbol. The wireless device may apply a permutation of bits (e.g., one or more nonequivalent permutations), such as (b_0, b_1, b_2, b_3) , (b_0, b_2, b_1, b_3) , or (b_0, b_2, b_3, b_1) .

[0096] The wireless device may apply a mapping function to map the 4-bit string (e.g., the permutation) to a 4-bit modulation symbol (e.g., constellation point) x . For example, for 256QAM (e.g., 16=PAM), a symbol-to-bit mapping function may be defined as $(c_0, c_1) \rightarrow (b_0, b_2, b_3, b_1)$, and a bit-to-constellation mapping function may be defined as $(b_0, b_2, b_3, b_1) \rightarrow 2(8b_0+4b_2+2b_3+b_1)-15$.

[0097] For 8-PAM, the wireless device may map three coded symbols c_0, c_1, c_2 to a bit string to obtain $(\phi(c_0), \phi(c_1), \phi(c_2))=(b_0, b_1, b_2, b_3, b_4, b_5)$. The wireless device may map (b_0, b_1, b_4) to first modulation symbol (e.g., constellation point) x_0 and (b_2, b_3, b_5) to a second modulation symbol (e.g., constellation point) x_1 . Additionally, or alternatively, the wireless device may map (b_0, b_4, b_1) to the first modulation symbol x_0 and (b_2, b_5, b_3) to the second modulation symbol x_1 . Additionally, or alternatively, the wireless device may map (b_4, b_0, b_1) to the first modulation symbol x_0 and (b_5, b_2, b_3) to the second modulation symbol x_1 . Such mappings may result in symbols with two different reliability levels and a bit-to-constellation mapping may be constant for each triplet of symbols **305**. Accordingly, such techniques may result in relatively less complex demodulation by a receiving wireless device.

[0098] In some examples, the wireless device may use one or more other bit-to-constellation mapping functions, such

as a gray-labeling function (e.g., $(c_0, c_1) \rightarrow \text{Gray}(b_0, b_2, b_3, b_1)$). Gray labeling for bit-independent coded modulation (BICM) may result in close bit-level capacities such that a reliability associated with demodulating the bits of the symbol **305-a** may be relatively close to a reliability associated with demodulating the bits of the symbol **305-b**. In some examples, a bit-level gray labeling may also result in a symbol-level gray mapping.

[0099] In some examples, to select the mapping procedure **310**, the wireless device may determine a gain of one or more mapping procedures **310**. The wireless device may accordingly select a mapping procedure **310** with a gain that is relatively higher than a gain of one or more other mapping procedures **310**. For example, a bit-to-constellation mapping **315-b** with a mapping function that maps the permutation (b_0, b_2, b_3, b_1) to a symbol x (e.g., as illustrated with reference to FIG. 3) may have a relatively higher gain than a bit-to-constellation mapping **315-b** with a mapping function that maps the permutation (b_0, b_1, b_2, b_3) to the symbol x . Additionally, the wireless device may select a mapping procedure **310** to result in similar bit-level or symbol-level capacities (e.g., and therefore similar demodulation reliability). For example, two mapping functions with same constellation subsets that correspond to different values of c_0 with a fixed value of c_1 up to a permutation may have similar (e.g., identical) demodulation performance for c_0 (e.g., and vice-versa).

[0100] FIG. 4 shows an example of a process flow **400** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The process flow **400** may implement or may be implemented by aspects of the wireless communications system **100**, the wireless communications system **200**, or the block diagram **300**. For example, the process flow **400** may be implemented by one or more wireless devices **402**, which may be examples of UEs **115** or network entities **105** as described with reference to FIG. 1.

[0101] In the following description of the process flow **400**, the operations between the wireless device **402-a** and the wireless device **402-b** may occur in a different order than the example order shown and, in some examples, may be performed by one or more different devices other than those shown as examples. Some operations also may be omitted from the process flow **400**, and other operations may be added to the process flow **400**. Further, although some operations or signaling may be shown to occur at different times for discussion purposes, these operations may actually occur at the same time.

[0102] At **405**, the wireless device **402-b** may perform an encoding procedure by inputting each of a set of symbols into one or more respective channels of a polar encoder. The set of information symbols may include one or more information symbols (e.g., with two or more information bits per symbol) and one or more frozen symbols (e.g., with two or more frozen bits per symbol). Each respective encoded information symbol obtained from the polar encoder may include at least two encoded information bits.

[0103] At **410**, the wireless device **402-b** may perform a mapping procedure on the set of encoded symbols (e.g., including the one or more encoded information symbols) to map the set of encoded symbols to one or more constellation points for transmission of the set of encoded symbols. The one or more constellation points may be associated with a modulation scheme (e.g., 256QAM, 16QAM, and the like).

In some examples, the wireless device **402-b** may select a mapping function associated with the mapping procedure (e.g., based on one or more of a gain of the mapping function, an error rate of the mapping function, a SNR of the mapping function, a reliability of decoding associated with the mapping function).

[0104] In some examples, the mapping procedure may include inputting each encoded symbol of the set of encoded symbols into a mapping function to map each encoded symbol to a respective constellation point (e.g., using a symbol-level gray labeling function or another mapping function). The wireless device **402-b** may map each bit of a first encoded symbol of the set of encoded symbols to a set of MSBs of a first constellation point and each bit of a second encoded symbol of the set of encoded symbols to a set of LSBs of the first constellation point, or may map the respective bits of each symbol to a different bit of the first constellation point.

[0105] Additionally, or alternatively, the mapping procedure may include converting the set of encoded symbols into a string of encoded bits (e.g., including the encoded information bits). The wireless device **402-b** may accordingly map each encoded bit of the string of encoded bits into a mapping function to map each encoded bit to a respective constellation point. For example, the wireless device **402-b** may map a first plurality of bits (e.g., information bits) of the string of encoded bits to a first constellation point and a second plurality of bits (e.g., information bits) of the string of encoded bits to a second constellation point. The mapping function may be a bit-level gray labeling function or another mapping function. In some examples, one or more reliability levels associated with the mapped set of encoded symbols may be below a threshold reliability level.

[0106] At **415**, the wireless device **402-b** may transmit a message to a wireless device **402-a** including the mapped set of encoded symbols (e.g., including the mapped set of encoded information symbols). The wireless device **402-a** may receive the message from the wireless device **402-b** (e.g., via a channel between the wireless device **402-a** and the wireless device **402-b**). The message may be a control message, a data message, or another type of message.

[0107] At **420**, the wireless device **402-a** may perform a demapping procedure on the mapped set of encoded symbols received via the message. The demapping procedure may demap the mapped set of encoded symbols from the constellation points associated with the modulation scheme. Each demapped encoded symbol of the set of encoded symbols may include two encoded bits (e.g., encoded information bits).

[0108] In some examples, performing the demapping procedure may include inputting each mapped encoded symbol into a demapping function to demap each symbol from a respective constellation point. In some examples, performing the demapping procedure may include inputting each of the mapped set of encoded symbols into a demapping function to obtain one or more strings of encoded bits, and converting the one or more strings of encoded bits into the set of encoded symbols. The demapping function may be a symbol-level max-log-map demapper.

[0109] In some examples, the wireless device **402-a** may obtain a LPMF associated with a respective one of each of the set of encoded symbols (e.g., including the set of

encoded information symbols). Each LMPF may be based on a plurality of probabilities of possible values of the respective encoded symbol.

[0110] At 425, the wireless device 402-a may perform an encoding procedure by inputting each of the encoded set of symbols into one or more respective channels of a polar decoder. The polar decoder may output a set of symbols, including the set of information symbols (e.g., with two or more information bits per symbol) and the set of frozen symbols (e.g., with two or more frozen bits per symbol).

[0111] FIG. 5 shows a block diagram 500 of a device 505 that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The device 505 may be an example of aspects of a UE 115 as described herein. The device 505 may include a receiver 510, a transmitter 515, and a communications manager 520. The device 505, or one or more components of the device 505 (e.g., the receiver 510, the transmitter 515, the communications manager 520), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0112] The receiver 510 may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to modulation schemes for nonbinary polar coding). Information may be passed on to other components of the device 505. The receiver 510 may utilize a single antenna or a set of multiple antennas.

[0113] The transmitter 515 may provide a means for transmitting signals generated by other components of the device 505. For example, the transmitter 515 may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to modulation schemes for nonbinary polar coding). In some examples, the transmitter 515 may be co-located with a receiver 510 in a transceiver module. The transmitter 515 may utilize a single antenna or a set of multiple antennas.

[0114] The communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be examples of means for performing various aspects of modulation schemes for nonbinary polar coding as described herein. For example, the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0115] In some examples, the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include at least one of a processor, a digital signal processor (DSP), a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some

examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0116] Additionally, or alternatively, the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0117] In some examples, the communications manager 520 may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver 510, the transmitter 515, or both. For example, the communications manager 520 may receive information from the receiver 510, send information to the transmitter 515, or be integrated in combination with the receiver 510, the transmitter 515, or both to obtain information, output information, or perform various other operations as described herein.

[0118] The communications manager 520 may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager 520 is capable of, configured to, or operable to support a means for inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The communications manager 520 is capable of, configured to, or operable to support a means for performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The communications manager 520 is capable of, configured to, or operable to support a means for transmitting a message including the mapped set of encoded information symbols.

[0119] Additionally, or alternatively, the communications manager 520 may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager 520 is capable of, configured to, or operable to support a means for receiving a message including a mapped set of encoded information symbols. The communications manager 520 is capable of, configured to, or operable to support a means for performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set

of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The communications manager 520 is capable of, configured to, or operable to support a means for inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0120] By including or configuring the communications manager 520 in accordance with examples as described herein, the device 505 (e.g., at least one processor controlling or otherwise coupled with the receiver 510, the transmitter 515, the communications manager 520, or a combination thereof) may support techniques for mapping and demapping symbols that are encoded using a nonbinary polar coding scheme, which may allow for more efficient utilization of communication resources.

[0121] FIG. 6 shows a block diagram 600 of a device 605 that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The device 605 may be an example of aspects of a device 505 or a UE 115 as described herein. The device 605 may include a receiver 610, a transmitter 615, and a communications manager 620. The device 605, or one or more components of the device 605 (e.g., the receiver 610, the transmitter 615, the communications manager 620), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0122] The receiver 610 may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to modulation schemes for nonbinary polar coding). Information may be passed on to other components of the device 605. The receiver 610 may utilize a single antenna or a set of multiple antennas.

[0123] The transmitter 615 may provide a means for transmitting signals generated by other components of the device 605. For example, the transmitter 615 may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to modulation schemes for nonbinary polar coding). In some examples, the transmitter 615 may be co-located with a receiver 610 in a transceiver module. The transmitter 615 may utilize a single antenna or a set of multiple antennas.

[0124] The device 605, or various components thereof, may be an example of means for performing various aspects of modulation schemes for nonbinary polar coding as described herein. For example, the communications manager 620 may include a nonbinary polar coding manager 625, a modulation manager 630, a message transmission manager 635, a message reception manager 640, a demodulation manager 645, a decoding manager 650, or any combination thereof. The communications manager 620 may be an example of aspects of a communications manager 520 as described herein. In some examples, the communications manager 620, or various components thereof, may be con-

figured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver 610, the transmitter 615, or both. For example, the communications manager 620 may receive information from the receiver 610, send information to the transmitter 615, or be integrated in combination with the receiver 610, the transmitter 615, or both to obtain information, output information, or perform various other operations as described herein.

[0125] The communications manager 620 may support wireless communications in accordance with examples as disclosed herein. The nonbinary polar coding manager 625 is capable of, configured to, or operable to support a means for inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The modulation manager 630 is capable of, configured to, or operable to support a means for performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The message transmission manager 635 is capable of, configured to, or operable to support a means for transmitting a message including the mapped set of encoded information symbols.

[0126] Additionally, or alternatively, the communications manager 620 may support wireless communications in accordance with examples as disclosed herein. The message reception manager 640 is capable of, configured to, or operable to support a means for receiving a message including a mapped set of encoded information symbols. The demodulation manager 645 is capable of, configured to, or operable to support a means for performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The decoding manager 650 is capable of, configured to, or operable to support a means for inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0127] FIG. 7 shows a block diagram 700 of a communications manager 720 that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The communications manager 720 may be an example of aspects of a communications manager 520, a communications manager 620, or both, as described herein. The communications manager 720, or various components thereof, may be an example of means for performing various aspects of modulation schemes for nonbinary polar coding as described herein. For example, the communications manager 720 may include a nonbinary

polar coding manager 725, a modulation manager 730, a message transmission manager 735, a message reception manager 740, a demodulation manager 745, a decoding manager 750, an information bit conversion manager 755, an information symbol conversion manager 760, an LPMF manager 765, or any combination thereof. Each of these components, or components or subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0128] The communications manager 720 may support wireless communications in accordance with examples as disclosed herein. The nonbinary polar coding manager 725 is capable of, configured to, or operable to support a means for inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The modulation manager 730 is capable of, configured to, or operable to support a means for performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The message transmission manager 735 is capable of, configured to, or operable to support a means for transmitting a message including the mapped set of encoded information symbols.

[0129] In some examples, to support performing the mapping procedure, the modulation manager 730 is capable of, configured to, or operable to support a means for inputting each encoded information symbol of the set of encoded information symbols into a mapping function to map each encoded information symbol to a respective constellation point of the one or more constellation points.

[0130] In some examples, the mapping function is a symbol-level gray labeling function.

[0131] In some examples, to support performing the mapping procedure, the modulation manager 730 is capable of, configured to, or operable to support a means for mapping a first encoded information symbol of the set of encoded information symbols to a set of most significant bits of a first constellation point of the one or more constellation points. In some examples, to support performing the mapping procedure, the modulation manager 730 is capable of, configured to, or operable to support a means for mapping a second encoded information symbol of the set of encoded information symbols to a set of least significant bits of the first constellation point.

[0132] In some examples, to support performing the mapping procedure, the information bit conversion manager 755 is capable of, configured to, or operable to support a means for converting the set of encoded information symbols into a string of encoded information bits. In some examples, to support performing the mapping procedure, the modulation manager 730 is capable of, configured to, or operable to support a means for inputting each encoded information bit of the string of encoded information bits into a mapping function to map each encoded information bit to a respective constellation point of the one or more constellation points.

[0133] In some examples, to support performing the mapping procedure, the modulation manager 730 is capable of, configured to, or operable to support a means for mapping a first set of multiple bits of the string of encoded information bits to a first constellation point of the one or more constellation points. In some examples, to support performing the mapping procedure, the modulation manager 730 is capable of, configured to, or operable to support a means for mapping a second set of multiple bits of the string of encoded information bits to a second constellation point of the one or more constellation points.

[0134] In some examples, one or more reliability levels associated with the mapped set of encoded information symbols are below a threshold reliability.

[0135] In some examples, the mapping function is a bit-level gray labeling function.

[0136] In some examples, to support inputting the one or more information symbols into the one or more respective channels of the polar encoder, the nonbinary polar coding manager 725 is capable of, configured to, or operable to support a means for inputting each respective one of the set of information symbols into a respective channel of the polar encoder.

[0137] In some examples, the modulation manager 730 is capable of, configured to, or operable to support a means for selecting a mapping function associated with the mapping procedure based on a gain associated with the mapping function, an error rate associated with the mapping function, a signal-to-noise ratio associated with a wireless channel, or any combination thereof.

[0138] Additionally, or alternatively, the communications manager 720 may support wireless communications in accordance with examples as disclosed herein. The message reception manager 740 is capable of, configured to, or operable to support a means for receiving a message including a mapped set of encoded information symbols. The demodulation manager 745 is capable of, configured to, or operable to support a means for performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The decoding manager 750 is capable of, configured to, or operable to support a means for inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0139] In some examples, to support performing the demapping procedure, the demodulation manager 745 is capable of, configured to, or operable to support a means for inputting each mapped encoded information symbol of the mapped set of encoded information symbols into a demapping function to demap each mapped encoded information symbol of the mapped set of encoded information symbols from a respective constellation point of one or more constellation points.

[0140] In some examples, to support performing the demapping procedure, the demodulation manager 745 is capable of, configured to, or operable to support a means for

inputting each of the mapped set of encoded information symbols into a demapping function to obtain one or more strings of encoded information bits associated with the mapped set of encoded information bits. In some examples, to support performing the demapping procedure, the information symbol conversion manager **760** is capable of, configured to, or operable to support a means for converting the one or more strings of encoded information bits into the set of encoded information symbols.

[0141] In some examples, to support performing the demapping procedure, the demodulation manager **745** is capable of, configured to, or operable to support a means for inputting each of the mapped set of encoded information symbols into a symbol-level max-log-map demapper.

[0142] In some examples, the LPMF manager **765** is capable of, configured to, or operable to support a means for obtaining a respective logarithmic probability mass function associated with each of the set of information symbols.

[0143] In some examples, each respective logarithmic probability mass function is based on a set of multiple probabilities associated with possible values of a respective one of the set of information symbols.

[0144] In some examples, to support inputting the one or more encoded information symbols into the one or more respective channels of the polar decoder, the decoding manager **750** is capable of, configured to, or operable to support a means for inputting each respective one of the set of encoded information symbols into a respective channel of the polar decoder.

[0145] FIG. 8 shows a diagram of a system **800** including a device **805** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The device **805** may be an example of or include components of a device **505**, a device **605**, or a UE **115** as described herein. The device **805** may communicate (e.g., wirelessly) with one or more other devices (e.g., network entities **105**, UEs **115**, or a combination thereof). The device **805** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **820**, an input/output (I/O) controller, such as an I/O controller **810**, a transceiver **815**, one or more antennas **825**, at least one memory **830**, code **835**, and at least one processor **840**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **845**).

[0146] The I/O controller **810** may manage input and output signals for the device **805**. The I/O controller **810** may also manage peripherals not integrated into the device **805**. In some cases, the I/O controller **810** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **810** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **810** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **810** may be implemented as part of one or more processors, such as the at least one processor **840**. In some cases, a user may interact with the device **805** via the I/O controller **810** or via hardware components controlled by the I/O controller **810**.

[0147] In some cases, the device **805** may include a single antenna. However, in some other cases, the device **805** may have more than one antenna, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **815** may communicate bi-directionally via the one or more antennas **825** using wired or wireless links as described herein. For example, the transceiver **815** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **815** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **825** for transmission, and to demodulate packets received from the one or more antennas **825**. The transceiver **815**, or the transceiver **815** and one or more antennas **825**, may be an example of a transmitter **515**, a transmitter **615**, a receiver **510**, a receiver **610**, or any combination thereof or component thereof, as described herein.

[0148] The at least one memory **830** may include random access memory (RAM) and read-only memory (ROM). The at least one memory **830** may store computer-readable, computer-executable, or processor-executable code, such as the code **835**. The code **835** may include instructions that, when executed by the at least one processor **840**, cause the device **805** to perform various functions described herein. The code **835** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **835** may not be directly executable by the at least one processor **840** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory **830** may include, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0149] The at least one processor **840** may include one or more intelligent hardware devices (e.g., one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPU) (also referred to as neural network processors or deep learning processors (DLP)), one or more micro-controllers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor **840** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the at least one processor **840**. The at least one processor **840** may be configured to execute computer-readable instructions stored in a memory (e.g., the at least one memory **830**) to cause the device **805** to perform various functions (e.g., functions or tasks supporting modulation schemes for nonbinary polar coding). For example, the device **805** or a component of the device **805** may include at least one processor **840** and at least one memory **830** coupled with or to the at least one processor **840**, the at least one processor **840** and the at least one memory **830** configured to perform various functions described herein. In some examples, the at least one processor **840** may include multiple processors and the at least one memory **830** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or

collectively, be configured to perform various functions described herein. In some examples, the at least one processor **840** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **840**) and memory circuitry (which may include the at least one memory **830**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **840** or a processing system including the at least one processor **840** may be configured to, configurable to, or operable to cause the device **805** to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code **835** (e.g., processor-executable code) stored in the at least one memory **830** or otherwise, to perform one or more of the functions described herein.

[0150] The communications manager **820** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **820** is capable of, configured to, or operable to support a means for inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The communications manager **820** is capable of, configured to, or operable to support a means for performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The communications manager **820** is capable of, configured to, or operable to support a means for transmitting a message including the mapped set of encoded information symbols.

[0151] Additionally, or alternatively, the communications manager **820** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **820** is capable of, configured to, or operable to support a means for receiving a message including a mapped set of encoded information symbols. The communications manager **820** is capable of, configured to, or operable to support a means for performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The communications manager **820** is capable of, configured to, or operable to support a means for inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where

a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits.

[0152] By including or configuring the communications manager **820** in accordance with examples as described herein, the device **805** may support techniques for mapping and demapping symbols that are encoded using a nonbinary polar coding scheme, which may allow for improved communication reliability and reduced latency.

[0153] In some examples, the communications manager **820** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **815**, the one or more antennas **825**, or any combination thereof. Although the communications manager **820** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **820** may be supported by or performed by the at least one processor **840**, the at least one memory **830**, the code **835**, or any combination thereof. For example, the code **835** may include instructions executable by the at least one processor **840** to cause the device **805** to perform various aspects of modulation schemes for nonbinary polar coding as described herein, or the at least one processor **840** and the at least one memory **830** may be otherwise configured to, individually or collectively, perform or support such operations.

[0154] FIG. 9 shows a flowchart illustrating a method **900** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The operations of the method **900** may be implemented by a UE or its components as described herein. For example, the operations of the method **900** may be performed by a UE **115** as described with reference to FIGS. 1 through 8. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0155] At **905**, the method may include inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The operations of **905** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **905** may be performed by a nonbinary polar coding manager **725** as described with reference to FIG. 7.

[0156] At **910**, the method may include performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The operations of **910** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **910** may be performed by a modulation manager **730** as described with reference to FIG. 7.

[0157] At **915**, the method may include transmitting a message including the mapped set of encoded information

symbols. The operations of **915** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **915** may be performed by a message transmission manager **735** as described with reference to FIG. 7.

[0158] FIG. 10 shows a flowchart illustrating a method **1000** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The operations of the method **1000** may be implemented by a UE or its components as described herein. For example, the operations of the method **1000** may be performed by a UE **115** as described with reference to FIGS. 1 through 8. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0159] At **1005**, the method may include inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The operations of **1005** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1005** may be performed by a nonbinary polar coding manager **725** as described with reference to FIG. 7.

[0160] At **1010**, the method may include performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The operations of **1010** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1010** may be performed by a modulation manager **730** as described with reference to FIG. 7.

[0161] At **1015**, the method may include inputting each encoded information symbol of the set of encoded information symbols into a mapping function to map each encoded information symbol to a respective constellation point of the one or more constellation points. The operations of **1015** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1015** may be performed by a modulation manager **730** as described with reference to FIG. 7.

[0162] At **1020**, the method may include transmitting a message including the mapped set of encoded information symbols. The operations of **1020** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1020** may be performed by a message transmission manager **735** as described with reference to FIG. 7.

[0163] FIG. 11 shows a flowchart illustrating a method **1100** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The operations of the method **1100** may be implemented by a UE or its components as described herein. For example, the operations of the method **1100** may be performed by a UE **115** as described with reference to FIGS.

1 through 8. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0164] At **1105**, the method may include inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, where a respective information symbol input into a respective channel includes at least two information bits, and where a respective encoded information symbol obtained from the polar encoder includes at least two encoded information bits. The operations of **1105** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1105** may be performed by a nonbinary polar coding manager **725** as described with reference to FIG. 7.

[0165] At **1110**, the method may include performing a mapping procedure on the set of encoded information symbols, where the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols. The operations of **1110** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1110** may be performed by a modulation manager **730** as described with reference to FIG. 7.

[0166] At **1115**, the method may include converting the set of encoded information symbols into a string of encoded information bits. The operations of **1115** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1115** may be performed by an information bit conversion manager **755** as described with reference to FIG. 7.

[0167] At **1120**, the method may include inputting each encoded information bit of the string of encoded information bits into a mapping function to map each encoded information bit to a respective constellation point of the one or more constellation points. The operations of **1120** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1120** may be performed by a modulation manager **730** as described with reference to FIG. 7.

[0168] At **1125**, the method may include transmitting a message including the mapped set of encoded information symbols. The operations of **1125** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1125** may be performed by a message transmission manager **735** as described with reference to FIG. 7.

[0169] FIG. 12 shows a flowchart illustrating a method **1200** that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The operations of the method **1200** may be implemented by a UE or its components as described herein. For example, the operations of the method **1200** may be performed by a UE **115** as described with reference to FIGS. 1 through 8. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0170] At 1205, the method may include receiving a message including a mapped set of encoded information symbols. The operations of 1205 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1205 may be performed by a message reception manager 740 as described with reference to FIG. 7.

[0171] At 1210, the method may include performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The operations of 1210 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1210 may be performed by a demodulation manager 745 as described with reference to FIG. 7.

[0172] At 1215, the method may include inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits. The operations of 1215 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1215 may be performed by a decoding manager 750 as described with reference to FIG. 7.

[0173] FIG. 13 shows a flowchart illustrating a method 1300 that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The operations of the method 1300 may be implemented by a UE or its components as described herein. For example, the operations of the method 1300 may be performed by a UE 115 as described with reference to FIGS. 1 through 8. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0174] At 1305, the method may include receiving a message including a mapped set of encoded information symbols. The operations of 1305 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1305 may be performed by a message reception manager 740 as described with reference to FIG. 7.

[0175] At 1310, the method may include performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The operations of 1310 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1310 may be performed by a demodulation manager 745 as described with reference to FIG. 7.

[0176] At 1315, the method may include inputting each mapped encoded information symbol of the mapped set of encoded information symbols into a demapping function to demap each mapped encoded information symbol of the mapped set of encoded information symbols from a respec-

tive constellation point of one or more constellation points. The operations of 1315 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1315 may be performed by a demodulation manager 745 as described with reference to FIG. 7.

[0177] At 1320, the method may include inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits. The operations of 1320 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1320 may be performed by a decoding manager 750 as described with reference to FIG. 7.

[0178] FIG. 14 shows a flowchart illustrating a method 1400 that supports modulation schemes for nonbinary polar coding in accordance with one or more aspects of the present disclosure. The operations of the method 1400 may be implemented by a UE or its components as described herein. For example, the operations of the method 1400 may be performed by a UE 115 as described with reference to FIGS. 1 through 8. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0179] At 1405, the method may include receiving a message including a mapped set of encoded information symbols. The operations of 1405 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1405 may be performed by a message reception manager 740 as described with reference to FIG. 7.

[0180] At 1410, the method may include performing a demapping procedure on the mapped set of encoded information symbols, where the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols including at least two encoded information bits. The operations of 1410 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1410 may be performed by a demodulation manager 745 as described with reference to FIG. 7.

[0181] At 1415, the method may include inputting each of the mapped set of encoded information symbols into a demapping function to obtain one or more strings of encoded information bits associated with the mapped set of encoded information bits. The operations of 1415 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1415 may be performed by a demodulation manager 745 as described with reference to FIG. 7.

[0182] At 1420, the method may include converting the one or more strings of encoded information bits into the set of encoded information symbols. The operations of 1420 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1420 may be performed by an information symbol conversion manager 760 as described with reference to FIG. 7.

[0183] At 1425, the method may include inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, where a respective information symbol of the set of information symbols obtained from the polar decoder includes two information bits. The operations of 1425 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1425 may be performed by a decoding manager 750 as described with reference to FIG. 7.

[0184] The following provides an overview of aspects of the present disclosure:

[0185] Aspect 1: A method for wireless communications by a wireless device, comprising: inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, wherein a respective information symbol input into a respective channel comprises at least two information bits, and wherein a respective encoded information symbol obtained from the polar encoder comprises at least two encoded information bits; performing a mapping procedure on the set of encoded information symbols, wherein the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols; and transmitting a message comprising the mapped set of encoded information symbols.

[0186] Aspect 2: The method of aspect 1, wherein performing the mapping procedure comprises: inputting each encoded information symbol of the set of encoded information symbols into a mapping function to map each encoded information symbol to a respective constellation point of the one or more constellation points.

[0187] Aspect 3: The method of aspect 2, wherein the mapping function is a symbol-level gray labeling function.

[0188] Aspect 4: The method of aspect 2, wherein performing the mapping procedure comprises: mapping a first encoded information symbol of the set of encoded information symbols to a set of most significant bits of a first constellation point of the one or more constellation points; and mapping a second encoded information symbol of the set of encoded information symbols to a set of least significant bits of the first constellation point.

[0189] Aspect 5: The method of aspect 1, wherein performing the mapping procedure comprises: converting the set of encoded information symbols into a string of encoded information bits; and inputting each encoded information bit of the string of encoded information bits into a mapping function to map each encoded information bit to a respective constellation point of the one or more constellation points.

[0190] Aspect 6: The method of aspect 5, wherein performing the mapping procedure comprises: mapping a first plurality of bits of the string of encoded information bits to a first constellation point of the one or more constellation points; and mapping a second plurality of bits of the string of encoded information bits to a second constellation point of the one or more constellation points.

[0191] Aspect 7: The method of any of aspects 5 through 6, wherein one or more reliability levels associated with the mapped set of encoded information symbols are below a threshold reliability.

[0192] Aspect 8: The method of any of aspects 5 through 7, wherein the mapping function is a bit-level gray labeling function.

[0193] Aspect 9: The method of any of aspects 1 through 8, wherein inputting the one or more information symbols into the one or more respective channels of the polar encoder comprises: inputting each respective one of the set of information symbols into a respective channel of the polar encoder.

[0194] Aspect 10: The method of any of aspects 1 through 9, further comprising: selecting a mapping function associated with the mapping procedure based at least in part on a gain associated with the mapping function, an error rate associated with the mapping function, a signal-to-noise ratio associated with a wireless channel, or any combination thereof.

[0195] Aspect 11: A method for wireless communications at a wireless device, comprising: receiving a message comprising a mapped set of encoded information symbols; performing a demapping procedure on the mapped set of encoded information symbols, wherein the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols comprising at least two encoded information bits; and inputting one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, wherein a respective information symbol of the set of information symbols obtained from the polar decoder comprises two information bits.

[0196] Aspect 12: The method of aspect 11, wherein performing the demapping procedure comprises: inputting each mapped encoded information symbol of the mapped set of encoded information symbol into a demapping function to demap each mapped encoded information symbol of the mapped set of encoded information symbol from a respective constellation point of one or more constellation points.

[0197] Aspect 13: The method of aspect 11, wherein performing the demapping procedure comprises: inputting each of the mapped set of encoded information symbols into a demapping function to obtain one or more strings of encoded information bits associated with the mapped set of encoded information bits; and converting the one or more strings of encoded information bits into the set of encoded information symbols.

[0198] Aspect 14: The method of any of aspects 11 through 13, wherein performing the demapping procedure comprises: inputting each of the mapped set of encoded information symbols into a symbol-level max-log-map demapper.

[0199] Aspect 15: The method of any of aspects 11 through 14, further comprising: obtaining a respective logarithmic probability mass function associated with each of the set of information symbols.

[0200] Aspect 16: The method of aspect 15, wherein each respective logarithmic probability mass function is based at least in part on a plurality of probabilities associated with possible values of a respective one of the set of information symbols.

[0201] Aspect 17: The method of any of aspects 11 through 16, wherein inputting the one or more encoded information symbols into the one or more respective channels of the polar decoder comprises: inputting each respec-

tive one of the set of encoded information symbols into a respective channel of the polar decoder.

[0202] Aspect 18: A wireless device for wireless communications, comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the wireless device to perform a method of any of aspects 1 through 10.

[0203] Aspect 19: A wireless device for wireless communications, comprising at least one means for performing a method of any of aspects 1 through 10.

[0204] Aspect 20: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by one or more processors to perform a method of any of aspects 1 through 10.

[0205] Aspect 21: A wireless device for wireless communications, comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the wireless device to perform a method of any of aspects 11 through 17.

[0206] Aspect 22: A wireless device for wireless communications, comprising at least one means for performing a method of any of aspects 11 through 17.

[0207] Aspect 23: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by one or more processors to perform a method of any of aspects 11 through 17.

[0208] It should be noted that the methods described herein describe possible implementations. The operations and the steps may be rearranged or otherwise modified and other implementations are possible. Further, aspects from two or more of the methods may be combined.

[0209] Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

[0210] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0211] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed using a general-purpose processor, a DSP, an ASIC, a CPU, a graphics processing unit (GPU), a neural processing unit (NPU), an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor but, in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be

implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Any functions or operations described herein as being capable of being performed by a processor may be performed by multiple processors that, individually or collectively, are capable of performing the described functions or operations.

[0212] The functions described herein may be implemented using hardware, software executed by a processor, firmware, or any combination thereof. If implemented using software executed by a processor, the functions may be stored as or transmitted using one or more instructions or code of a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0213] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one location to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disc, and Blu-ray disc. Disks may reproduce data magnetically, and discs may reproduce data optically using lasers. Combinations of the above are also included within the scope of computer-readable media. Any functions or operations described herein as being capable of being performed by a memory may be performed by multiple memories that, individually or collectively, are capable of performing the described functions or operations.

[0214] As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that

is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0215] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” and “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, the term “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” may refer to any or all of the one or more components. For example, a component introduced with the article “a” may be understood to mean “one or more components,” and referring to “the component” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.” Similarly, subsequent reference to a component introduced as “one or more components” using the terms “the” or “said” may refer to any or all of the one or more components. For example, referring to “the one or more components” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.”

[0216] The term “determine” or “determining” encompasses a variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database, or another data structure), ascertaining, and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data stored in memory), and the like. Also, “determining” can include resolving, obtaining, selecting, choosing, establishing, and other such similar actions.

[0217] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label or other subsequent reference label.

[0218] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some figures, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0219] The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A wireless device, comprising:

one or more memories storing processor-executable code; and

one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the wireless device to:

input one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, wherein a respective information symbol input into a respective channel comprises at least two information bits, and wherein a respective encoded information symbol obtained from the polar encoder comprises at least two encoded information bits;

perform a mapping procedure on the set of encoded information symbols, wherein the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols; and

transmit a message comprising the mapped set of encoded information symbols.

2. The wireless device of claim 1, wherein, to perform the mapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

input each encoded information symbol of the set of encoded information symbols into a mapping function to map each encoded information symbol to a respective constellation point of the one or more constellation points.

3. The wireless device of claim 2, wherein the mapping function is a symbol-level gray labeling function.

4. The wireless device of claim 2, wherein, to perform the mapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

map a first encoded information symbol of the set of encoded information symbols to a set of most significant bits of a first constellation point of the one or more constellation points; and

map a second encoded information symbol of the set of encoded information symbols to a set of least significant bits of the first constellation point.

5. The wireless device of claim 1, wherein, to perform the mapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

convert the set of encoded information symbols into a string of encoded information bits; and

input each encoded information bit of the string of encoded information bits into a mapping function to map each encoded information bit to a respective constellation point of the one or more constellation points.

6. The wireless device of claim 5, wherein, to perform the mapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

map a first plurality of bits of the string of encoded information bits to a first constellation point of the one or more constellation points; and

map a second plurality of bits of the string of encoded information bits to a second constellation point of the one or more constellation points.

7. The wireless device of claim 5, wherein one or more reliability levels associated with the mapped set of encoded information symbols are below a threshold reliability.

8. The wireless device of claim 5, wherein the mapping function is a bit-level gray labeling function.

9. The wireless device of claim 1, wherein, to input the one or more information symbols into the one or more respective channels of the polar encoder, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

input each respective one of the set of information symbols into a respective channel of the polar encoder.

10. The wireless device of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the wireless device to:

select a mapping function associated with the mapping procedure based at least in part on a gain associated with the mapping function, an error rate associated with the mapping function, a signal-to-noise ratio associated with a wireless channel, or any combination thereof.

11. A wireless device, comprising:

one or more memories storing processor-executable code; and

one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the wireless device to:

receive a message comprising a mapped set of encoded information symbols;

perform a demapping procedure on the mapped set of encoded information symbols, wherein the demapping procedure demaps the mapped set of encoded information symbols to obtain a set of encoded information symbols, a respective encoded information symbol of the set of encoded information symbols comprising at least two encoded information bits; and

input one or more encoded information symbols of the set of encoded information symbols into one or more respective channels of a polar decoder to obtain a set of information symbols, wherein a respective information symbol of the set of information symbols obtained from the polar decoder comprises two information bits.

12. The wireless device of claim 11, wherein, to perform the demapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

input each mapped encoded information symbol of the mapped set of encoded information symbols into a

demapping function to demap each mapped encoded information symbol of the mapped set of encoded information symbols from a respective constellation point of one or more constellation points.

13. The wireless device of claim 11, wherein, to perform the demapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

input each of the mapped set of encoded information symbols into a demapping function to obtain one or more strings of encoded information bits associated with the mapped set of encoded information bits; and convert the one or more strings of encoded information bits into the set of encoded information symbols.

14. The wireless device of claim 11, wherein, to perform the demapping procedure, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

input each of the mapped set of encoded information symbols into a symbol-level max-log-map demapper.

15. The wireless device of claim 11, wherein the one or more processors are individually or collectively further operable to execute the code to cause the wireless device to:

obtain a respective logarithmic probability mass function associated with each of the set of information symbols.

16. The wireless device of claim 15, wherein each respective logarithmic probability mass function is based at least in part on a plurality of probabilities associated with possible values of a respective one of the set of information symbols.

17. The wireless device of claim 11, wherein, to input the one or more encoded information symbols into the one or more respective channels of the polar decoder, the one or more processors are individually or collectively operable to execute the code to cause the wireless device to:

input each respective one of the set of encoded information symbols into a respective channel of the polar decoder.

18. A method for wireless communications by a wireless device, comprising:

inputting one or more information symbols of a set of information symbols into one or more respective channels of a polar encoder to obtain a set of encoded information symbols, wherein a respective information symbol input into a respective channel comprises at least two information bits, and wherein a respective encoded information symbol obtained from the polar encoder comprises at least two encoded information bits;

performing a mapping procedure on the set of encoded information symbols, wherein the mapping procedure maps the set of encoded information symbols to one or more constellation points associated with a modulation scheme for transmission of the set of encoded information symbols to obtain a mapped set of encoded information symbols; and

transmitting a message comprising the mapped set of encoded information symbols.

19. The method of claim 18, wherein performing the mapping procedure comprises:

inputting each encoded information symbol of the set of encoded information symbols into a mapping function to map each encoded information symbol to a respective constellation point of the one or more constellation points.

20. The method of claim **18**, wherein performing the mapping procedure comprises:

converting the set of encoded information symbols into a string of encoded information bits; and

inputting each encoded information bit of the string of encoded information bits into a mapping function to map each encoded information bit to a respective constellation point of the one or more constellation points.

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